

Anti-Biofouling and Healable Materials: Preparation, Mechanisms, and Biomedical Applications

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Preventing biofouling is challenging for a wide range of biomedical applications. Although materials with good anti-biofouling properties are reported, their antifouling functions are readily lost when detachments or scratches occur through physical damage and chemical degradation by water, oxygen, or possible catalytic ions in water containing reactive environments. Consequently, it is important to develop durable anti-biofouling materials with healing abilities. This review summarizes the important developments in five kinds of anti-biofouling materials categorized according to healing mechanisms and material properties in the recent five years, namely structural rearrangement-, reaction-, ionic bond-, and hydrogel-based healing materials, as well as other healable antifouling materials. The healing mechanisms and potential biomedical applications of these healable anti-biofouling materials are emphasized.

1. Introduction

Biofouling, generally initiated by nonspecific adsorption of proteins, cells, or microorganisms on the surfaces of substrates, is a ubiquitous and persistent challenge for various applications including but not limited to biomedical devices,^[1] biosensors,^[2] membrane systems,^[3] and marine coatings.^[4] For example, biofouling of implanted biomaterials can cause an adverse effect on wound healing process, which has posed significant danger to human health. Because of the undesired effects, the fouling mechanisms and anti-biofouling strategies have been widely studied. The increasing number of publications shows the intensification of researches on new generations of anti-biofouling technologies. These studies show that the effective antifouling strategy is to prevent the initial stage of biofouling that is preventing biofoulants from attachment on the surfaces and degrading or killing them.^[5]

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Recently, reviews on the preparation and application of antifouling surfaces have been published, such as synthetic polymers (polyethylene glycol (PEG) and zwitterionic polymers),^[6] bioinspired microstructured surfaces,^[7] sol-gel and foul-releasing coatings^[8] in various applications.^[9,10] These references shed light on the typical types of anti-biofouling materials or anti-biofouling mechanisms.

Though good antifouling properties have been reported, most antifouling materials are soft and easy to be mechanically damaged such as inevitable scratches and partial exfoliation after long-term use, leading to the loss of their

antifouling functions. Accordingly, healing of the damage is very important to improve the stability of anti-biofouling materials in long-term use.

Healable materials are rapidly emerging research topics inspired by the healing processes of natural organisms. Based on the healing characteristics, they can be divided into autonomous and nonautonomous materials. Autonomous materials, also known as self-healing materials, are able to heal themselves without any stimuli, and self-healing is occurred simultaneously upon mechanical damage to the materials. In contrast, nonautonomous materials do not have the characteristics of self-healing, but are still healable. They could repair themselves upon heat, light, pressure, mechanical stress, or other external stimuli. Additionally, self-healing includes intrinsic and extrinsic strategies based on the number of healing events. Self-healing materials which focused on microcapsule or microtubule release of healing agents and repolymerization to achieve repairing belong to extrinsic healing materials. When cracks or damages occurred, the encapsulated healing agents were released to trigger the repairing by polymerization or chemical reactions. However, these materials were only able to be healed once or twice. Conversely, intrinsic self-healing is repeatable and self-healing materials are designed based on reversible crosslinks, either via dynamic covalent bonds or physical supramolecular (noncovalent) interactions. Specially, the self-healing or healing ability prolongs not only the lifetime of materials but also the original functions.^[11]

Hence, the introduction of self-healing or healable ability into anti-biofouling materials is important to improve their service life as well as efficiency, which has attracted broad attention. This review will therefore provide an overview of anti-biofouling materials with healing ability in recent years.

We focus on the “functional” self-healing, i.e., the recovery of anti-biofouling properties. Five types of anti-biofouling and healable materials are discussed in detail based on the healing mechanisms and material properties. Potentials of these anti-fouling mechanisms in various biomedical applications and some future perspectives are proposed.

2. Anti-Biofouling and Healable Materials

The general requirements of designing antifouling and healable materials are introduced in detail in this section based on healing mechanisms and the materials used. Three types of anti-biofouling and healable materials, namely, structural rearrangement-, reaction-, and ionic bond-based antifouling healable materials were classified based on the healing mechanisms. Additionally, hydrogel-based antifouling self-healing materials were listed as a specific class of materials. Each item was discussed and summarized in **Table 1** with several recent examples reported by various research groups. In addition, some other antifouling healable materials were also introduced briefly.

Generally, the healing process is characterized by morphology-based methods such as rheology, tensile tests, and cut-/scratch-healing tests with the assistance of scanning electron microscopy (SEM), optical microscope images, and atomic force microscopy (AFM), which also quantify the healing efficiency. Infrared (IR) spectroscopy or linear Raman spectra can monitor the formation or cleavage of chemical bonds and supramolecular interactions in the healing process. Recently, Popp and co-workers proposed that coherent anti-Stokes Raman scattering, a nonlinear Raman variant can be used as molecular characterization of healing process. It acquires morphological and molecular information simultaneously and thus offers a different observation from conventional morphological-based methods.^[12]

The anti-biofouling properties are usually evaluated by wettability measurements as well as the resistance ability to proteins, cells, bacteria, algae, and other microorganisms. In the wettability measurements, contact angle (CA) and contact angle hysteresis (CAH) experiments are used to consider the relationship between antifouling and wettability. In the protein adsorption test, bovine serum albumin (BSA) is usually used as a model protein, and the protein adhesion can be easily observed from fluorescence microscopy images after incubating in fluorescein isothiocyanate-labeled BSA (BSA-FITC) solution for a certain time. Gram-negative and gram-positive bacteria such as *Escherichia coli* and *Bacillus subtilis* were chosen typically in the bacteria resistance ability tests of the materials. Moreover, after treating by biofoulants, the morphology changes of these surfaces can be evaluated by tapping mode AFM measurements and SEM.^[13]

2.1. Structural Rearrangement-Based Materials

Structural rearrangement is a feasible approach to repair cracks or scratches via exposing the damaged area to solvents or heat, leading to re-entanglement and diffusion of the polymer chains. This mechanism is particularly used in antifouling coatings. The antifouling agents are filled both on the surface and



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in matrix and can diffuse from the host to surface once the surface is damaged, allowing the regain of antifouling properties in the cracks. This mechanism will remind readers the extrinsic self-healing systems, in which micrometer-sized capsules load healing agents and catalysts, respectively. When damage happens, the microcapsules (MCs) near the damaged sites also crack and release the healing agents and catalysts to trigger the chemical reactions for healing.

PEG and derivatives are the most widely used polymers for anti-biofouling applications, and immobilizing PEG is a commonly used method to impart antifouling ability to a surface.^[14] Conventionally, PEG can be attached to surfaces through physical or chemical adsorption, covalent attachment, and block/graft copolymerization.^[15] However, the degradation and detachment of grafted polymer might cause the loss of antifouling functions. To solve this problem, Minko and co-workers^[16] reported a 3D polymer-grafting approach to introduce poly(ethylene oxide) (PEO) chains on and inside the polymer films (**Figure 1**). Briefly, poly(2-vinyl pyridine) (P2VP) film was crosslinked by quaternization reaction, and halogen-terminated PEO was grafted to the P2VP films. Then, 3D grafting of PEO was achieved as PEO chains penetrated into P2VP network, which was demonstrated by ellipsometry and X-ray reflectivity measurements. The quaternization degree of the P2VP films and the PEO molecular weight significantly

Table 1. Material compositions, properties, healing mechanisms, and potential applications of the four kinds of healable and anti-biofouling materials.

Categories	Materials	Properties	Healable mechanism	Potential applications	Ref.
Structural rearrangement-based healable and anti-biofouling materials	PEO-grafted P2VP films	Antifouling, self-healing, stimuli-responsive	Free energy change-driven rearrangement of the grafted PEO chains	Biointerfaces in biosensors, diagnostic devices, drug delivery, cell culturing, separation filters	[16]
	3D grafting of triblock copolymer on PU coatings	Self-repairing antiprotein/blood platelet properties	Chemical potential gradient-driven replenishing of lost hydrophilic polymeric chains	Biomedical implants and devices, ships, marine equipment	[17]
	PMAF17-grafted nanostructured silicon surface	Superhydrophobicity, protein resistance, self-healing upon heat	Low surface energy-driven reorientation and reformation of the brush polymer	Long-term antifouling coatings applied in harsh environments	[20]
	Hydrophilic copolymer-grafted microgel spheres	Self-healing, superoleophobic, oil-repellent, protein resistance, thermosensitive	Microgel sphere swelling caused recovery of topography and hydrophilicity	Marine/aqueous environment	[22]
	FAS/PVDF-HFP/silica NPs-coated fabrics	Self-healing, superamphiphobicity, stable to strong acid/base, ozone, and boiling treatments	Surface free energy changing-driven migration of the fluorinated alkyl chains to the coating surface	Protective clothing	[24]
	Lyophobic Teflon NPs/fluorocarbon surfactant/fluorinated alkyl silane coating	Self-healing, superamphiphobicity, resistance to abrasion, washing, boiling, acid/base treatment	Migration of fluorinated alkyl silane under high temperature	Waterborne coating systems	[25]
	HPU/SRGO nanocomposite	Self-healable, shape-memory, microbial inhibitory effect	Shape memory of HPU and energy absorbing capability of SRGO-assisted melting, diffusion, and rearrangement of the soft segment of HPU to crack place	Various antimicrobial applications	[28]
Reaction-based healable and anti-biofouling materials	F-POSS-/Ag NP-/PEI film-coated cotton fabrics	Color fastness, self-healing, antibacterial, superhydrophobic	Increase of surface energy-driven migration of the embedded F-POSS molecules to the surface	Protective clothing in moist and insanitary environments	[31]
	Poly(carboxybetaine) ester-PU	Long-term stable and self-recoverable anti-biofouling properties	Self-regeneration of zwitterionic polymer brush coating by hydrolysis	Coating for marine and medical devices	[35]
	PTCBE elastomer	High bulk mechanical strength, elastomeric, nonfouling surface	Hydrolysis of "hidden zwitterion"	Biomedical and engineering applications	[36]
	SMA and bPEI coating on PP-g-MA support	Antimicrobial, self-healing upon heating	Regeneration of the imide bonds	Reducing microbial cross-contamination in food/ biomedical industries	[38]
Ionic bond-based healable and anti-biofouling materials	Chemical-filled PUF MCs	Superhydrophobicity, anticorrosion, antifouling	Formation of polyurea via reaction between liquid healant released from damaged areas with moisture or water	Multifunctional smart coatings in environmental, energy, and biomedical fields	[39]
	Triclosan-(bPEI/PAA)*n films	Optically transparent, self-healing, antibacterial ability	Electrostatic and hydrogen bonding interaction-driven high flowable and recombinable ability of bPEI and PAA in water	Applications in optical and electrical devices as protective films	[41]
	Crosslinked PEGylated bPEI/HA films	Antifouling, self-healing	Reversible electrostatic and hydrogen bonding interaction-driven reformation of crosslinked bPEI-PEG/HA	Self-healing antifouling films	[42]
	(PAA/PEtOx-EI-7%)*n films	Antifouling abilities to bacteria, capable of healing deep and wide cuts	Reformation of the hydrogen-bonding interactions in the fractured surfaces	Healable antifouling films	[45]
	BC-based polyelectrolyte film	Antimicrobial, self-healable	Ionic interaction between NH_3^+ and $-\text{COO}^-$ groups	Drug delivery, wound dressing	[46]
	Zwitterionic polymer network coatings	Self-healing, anti-biofouling	Reformation of ionic bond between protonated ammonium and sulfonic groups	Long-term stable biosensors and anti-biofouling coatings	[48]
	Zwitterionic (DURA-Z) coating	Ultradurable, long-term biofilm resistance to bacteria and fungi	Formation of zwitterionic pairs via electrostatic interactions	Anti-biofilm coatings, medical device, and marine coatings	[49]

Table 1. Continued.

Categories	Materials	Properties	Healable mechanism	Potential applications	Ref.
Hydrogels	ABA triblock copolymer–self-assembled hydrogel	Self-healing, anti-biofouling, antimicrobial, injectable, thermosensitivity, low cytotoxicity	Thermoresponsive sol-to-gel transition through hydrogen bonding and aromatic interactions	Bioengineering	[55,56]
	PNAGA–PVTZ hydrogel	Good mechanical property, antibacterial, anti-inflammatory, heat-triggered healing	Reformation of hydrogen bonds at the broken interface	Injectable hydrogel scaffolds in biomedical and engineering fields	[58]
	Ln–PVA hydrogel	High mechanical strength, antibacterial activity, biocompatibility, photoluminescence	–	Bioimaging, biosensors	[59]
	PAA-grafted quaternized cellulose and PVA double network hydrogel	Self-healing, antibacterial, mechanical performance, shape memory	Ionic bonding between ferric ions and carboxylic groups, hydrogen bonding between the PVA molecules	Various biomedical applications	[61]
	CS–AT/PEG–DA and QCSP/PEGS–FA hydrogel	Self-healing, conductive, injectable, antibacterial	Mobility of polymer chains, dynamic Schiff base between amine and benzaldehyde groups	Cell delivery carriers for cardiac repair, bioactive wound dressings	[63,64]
	Enzyme-induced EPL-g-poly(NVP-co-NMA) hydrogels	Self-healing, antibacterial	Intrinsic hydrogen bonds, dynamic Schiff base triggered by enzyme coordination	Wound repair, artificial skin, tissue engineering	[65]
	BMOD/PEGMA/HEAA/META hydrogels	Antifouling, antibacterial, self-healing	Rebonding of the disulfide bonds via disulfide exchange reaction at the cut interface	Durable coating in long-term biomaterial applications	[67]
	CBMAA-3-co-HEMA hydrogels	Antifouling, self-healing, superior mechanical property	Zwitterionic comonomers	Biomedical applications	[68]
	Thiol/acetyl thioester groups–Ag NP–based hydrogels	Good mechanical properties, antifungal performance, self-recovery	Dynamic interactions between Ag NPs and thiol/acetyl thioester groups	Marine coating, biomaterial interfaces	[69]

affected the PEO grafting, as higher crosslinking degree resulted in smaller network mesh size and slower PEO chain mobility. The 3D polymer-grafting films showed long-lasting antifouling effect, thanks to their self-healing property that originated from the replacement of damaged PEO on the surface by those inside the polymeric network. Theoretical analysis indicated that when damages happened, the balance of the free energy of the PEO on surface and in the network changed. The changes in the free energy impelled the rearrangement of PEO chains inside the network to maintain them with a higher density on the surface and replenish the damaged PEO. However, the drawback of this method is the complex preparation process and limited film thickness of only a few tens of nanometers. Zhou and co-workers also reported a similar method to develop a triblock copolymer–grafted polyurethane (PU) coating that exhibited self-repairing antifouling properties against protein and blood platelet.^[17]

Superhydrophobic surfaces possess superior water-repelling ability with a water CA larger than 150°. Superhydrophobic surfaces with micro-/nanostructures inspired from natural surfaces, such as the lotus leaves and the compound eyes of insects, are widely used in self-cleaning, ice accumulation prevention, antifouling, anticorrosion, and other fields.^[18] To improve the stability and overcome fouling problems, Wang and Zuillhof reported some approaches by covalently grafting commercial fluoropolymer brushes (poly(2-perfluorooctylethyl methacrylate), PMAF17) onto an atomically flat Si (111) surface.^[19,20] For example, PMAF17 was grafted onto a nanostructured silicon

surface (etched by AgNO₃/HF) by surface-initiated atom transfer radical polymerization.^[20] This fluoropolymer brush–modified nanostructured silicon surface showed excellent antiprotein fouling property, superhydrophobicity, and low surface energy, which was apparent from the small sliding angle (SA) (6°) and low CAH (8°). In addition, the damaged nanostructured brush could be repaired by heating to 120 °C for 30 min and maintained excellent antifouling and superhydrophobicity even after 5 damage–repair cycles. They hypothesized that when the temperature was above the glass transition temperature (*T_g*) of the brush polymer, the polymer chain displayed sufficient mobility to adapt itself and the undamaged fluorinated tails came to the surface to recover the hydrophobicity and antifouling properties (**Figure 2**).

Underwater superoleophobicity materials have the potential applications in oil contamination repellence. Zhou and co-workers have reported several underwater superoleophobic strategies even with self-healing ability.^[21] Combined with anti-biofouling property, in 2016, they successfully fabricated a self-healing underwater oil-repellent and anti-biofouling coating for the first time.^[22] As shown in **Figure 3a**, microgel spheres were coated by SiO₂ to obtain the hierarchical microgel spheres. Then, a hydrophilic copolymer was grafted to the surfaces of hierarchical microgel spheres to obtain modified hierarchical microgel spheres (MHMSs). Because of the unique hierarchical microgel spheres and sufficient hydrophilic polymer chains, a layer of water molecules was formed on the surface of MHMS-based film to reduce the contact between oil and material surface in aqueous medium. Moreover, the grafted

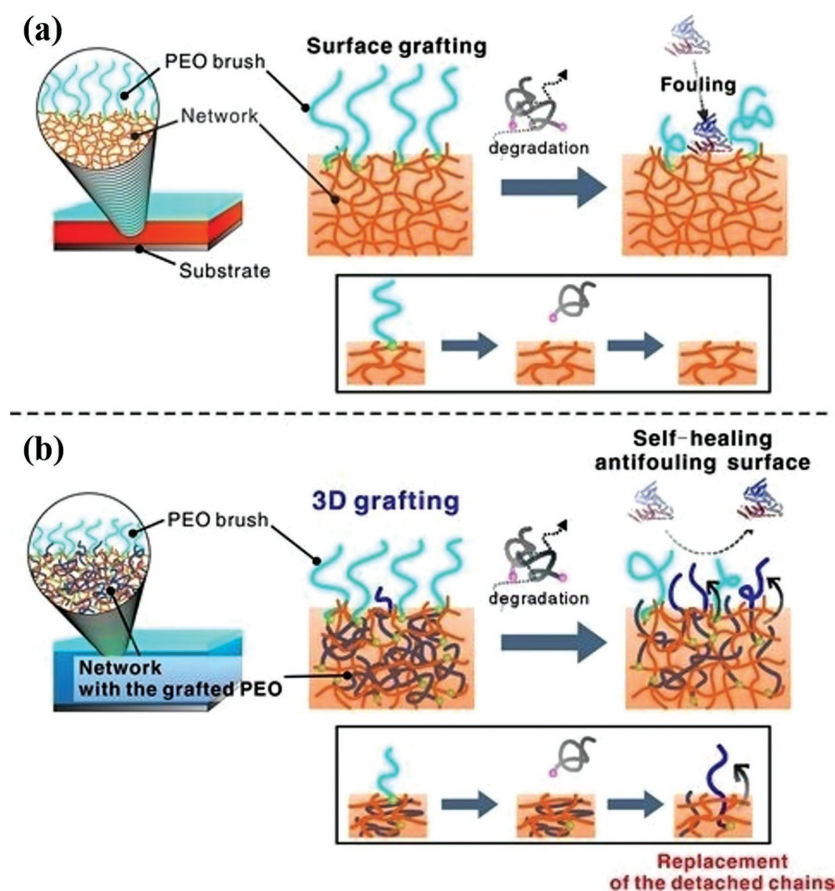


Figure 1. Schematic construction of antifouling films by PEO and P2VP: a) conventional approach, PEO was grafted onto the surface of P2VP film; b) 3D polymer grafting approach, PEO was grafted on and inside the P2VP film. All panels reproduced with permission.^[16] Copyright 2013, Wiley.

copolymer consisted of noncytotoxic 2-methacryloyloxyethyl phosphorylcholine, which also efficiently prevented the adhesion of marine organisms. The self-healing ability was due to the crosslinking effect between $-OH$ of MHMS and crosslinking agent, 1,6-hexamethylene diisocyanate trimer. When mechanically damaged, temperature-dependent swelling of the microgel spheres immersed in water helped the coating to recover its original topography and at the same time, the hydrophilic polymeric chains on the spheres replenished the lost hydrophilicity at the liquid/solid interface^[22] (Figure 3b).

Unlike superhydrophobic surfaces, superamphiphobic surfaces have both superior water-repellency and oil-repellency abilities, making them better candidates in applications of self-cleaning and antifouling.^[23] In 2013, Lin and co-workers reported a self-healing and superhydrophobic fabric prepared by a two-step wet-chemistry coating technique using commercially available materials.^[24] As shown in **Figure 4**, the fluoroalkyl silane (FAS) and poly(vinylidene fluoride-hexafluoropropylene) (PVDF-HFP) and silica nanoparticle-coated fabrics showed super-repellence to liquid drops even after 1 h. Specially, the coating is very stable to resist not only physical laundry and abrasion, but also strong acids/bases, ozone, and boiling treatments. After being damaged chemically

by plasma treatment, the amphiphobicity was reduced due to the introduction of polar groups to the surfaces. However, the superamphiphobicity could recover by a simple room temperature aging process and this process can be accelerated by heating treatment. The possible reason is that the coating has a low T_g and the molecular migration mobility in the coating is high, and the fluorinated alkyl chains tend to migrate to the coating surface in order to minimize the surface free energy. This molecular migration can be improved in high temperature, resulting in an increased healing speed. Moreover, this self-healing process was repeatable for many cycles. With robust, self-healing, and superamphiphobic surfaces, these reported fabrics showed great promise in the development of protective clothing.

Though simple and effective, the use of organic solvents in wet-chemistry coating techniques still limits their practical applications. To solve this problem, in 2017, Lin and co-workers reported a waterborne coating system by dispersing lyophobic Teflon nanoparticles, fluorocarbon surfactant (DuPont Zonyl321), and fluorinated alkyl silane in water.^[25] This system was applied to several substrates including fabrics, sponges, woods, glass, and metals. All the substrates showed superamphiphobicity with low CAH after being coated. The coating could stand physical abrasion, washing, boiling in water, as well as acid/base treatment, and showed self-healing ability by heating after plasma treatment. The heat-induced healing ability was

also ascribed to the migration of fluorinated alkyl silane under high temperature.

Some shape-memory polymers also have the application potential in the self-healing process as they can recover their original shapes from temporary deformed shapes upon external stimuli.^[26] Shape-memory polymers can be considered to be composed of soft and hard segments. The soft segment is generally an amorphous part or crystalline zone that hinders the polymer chain to reform a coil-like structure and thus can fix the shape-memory polymers in the deformed shape. The hard segment is crosslinked structure from entanglement of the polymer chains that stores entropy energy and controls the permanent shape of the polymers. Softening the oriented polymer chains by heating can regain the coiled conformation of the molecular chains. Therefore, the shape recovery of shape-memory polymers is driven by elastic energy generated during deformation.^[27,28] Many polymers possess shape-memory effect, among them PUs are the most promising due to their high recoverable strain, wide transition temperature range, inherent soft-hard segments, and tunable physical properties.^[27] In 2015, Karak and co-workers reported hyperbranched PU (HPU)/sulfur nanoparticle-decorated reduced graphene oxide (SRGO) nanocomposites.^[28] As shown in **Figure 5**, both

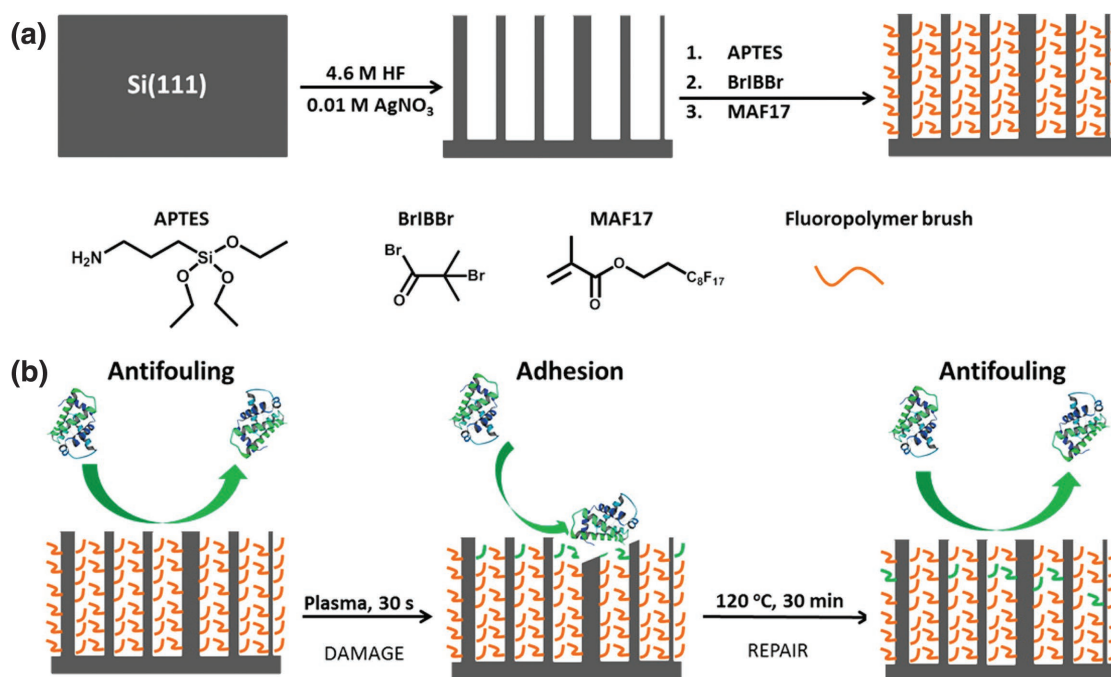


Figure 2. a) Schematic representation of fluoropolymer brushes (PMF17) onto an atomically flat Si surface; b) schematic illustration of the protein adhesion behavior on the freshly prepared, damaged, and repaired fluoropolymer brush-modified nanostructured silicon surface. All panels reproduced with permission.^[20] Copyright 2016, American Chemical Society.

HPU and HPU/SRGO exhibited shape memory properties under direct sunlight, as the soft segment could be melted by absorbing external energy, while the hard segment helped to retain their original shape. Compared with pure HPU matrix, the incorporation of SRGO exhibited tremendous enhancement of mechanical properties and inhibition of the growth

of *Staphylococcus aureus*, *E. coli*, and *Candida albicans*. In contrast to pure HPU which only possessed shape memory, HPU/SRGO also exhibited good healing ability. Supported by the shape memory property of HPU, the healing ability of HPU/SRGO came from the good energy absorbing capability and superior thermal conductivity of SRGO, which could absorb

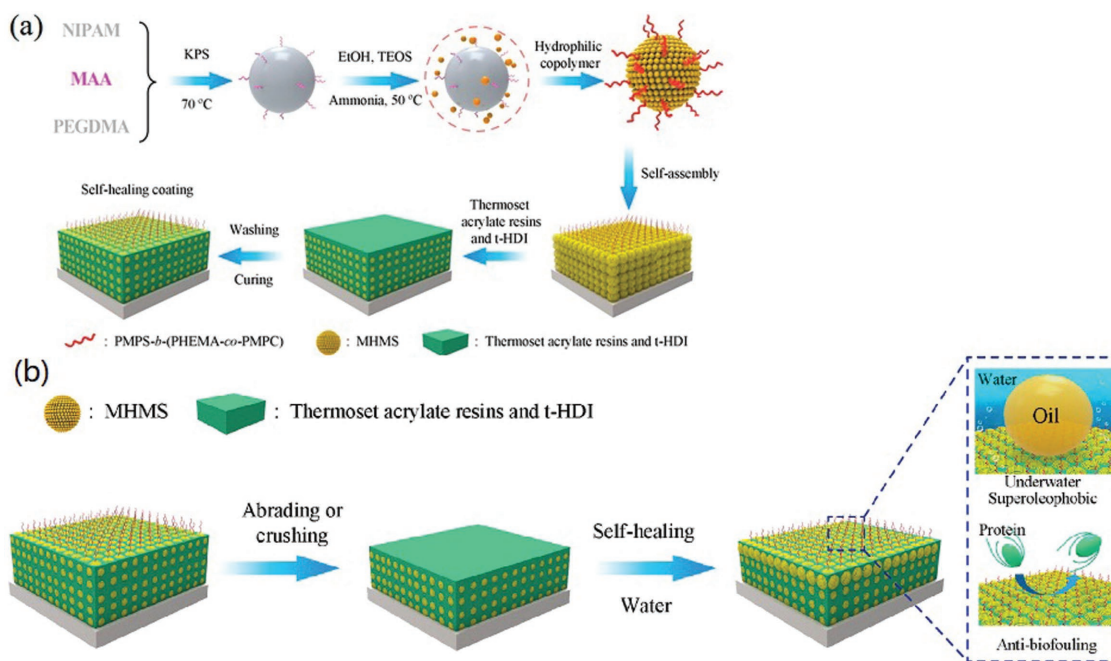


Figure 3. a) Schematic illustration for the preparation of MHMS-based coating; b) working principle of the self-repairing underwater superoleophobic/antifouling coating. All panels reproduced with permission.^[22] Copyright 2016, American Chemical Society.

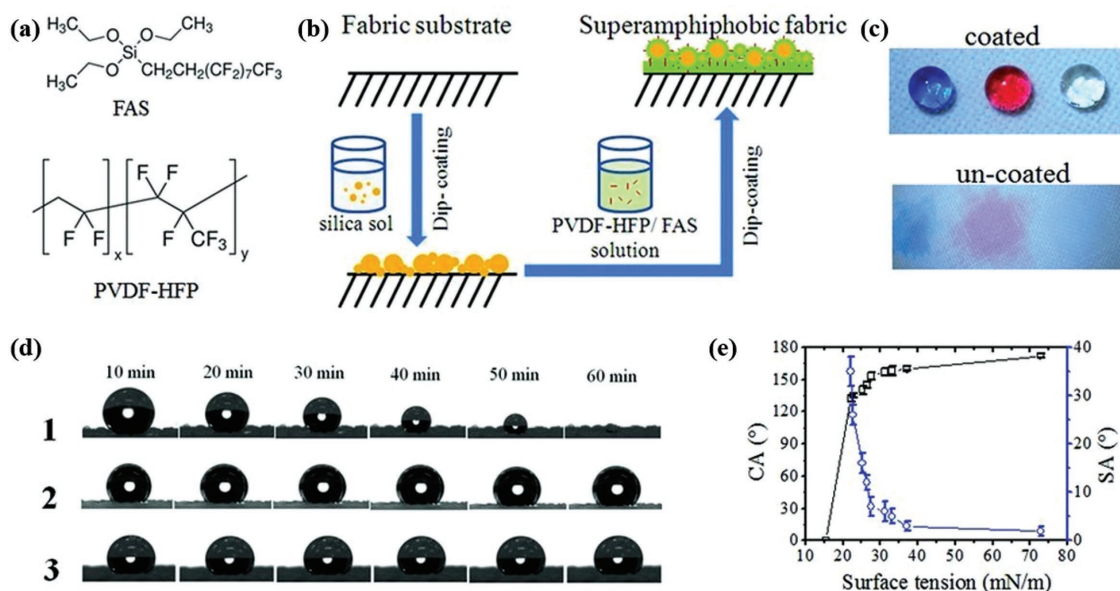


Figure 4. a) Chemical structure of FAS and PVDF-HFP; b) schematic coating procedure; c) photographs of water (blue), hexadecane (red), and soybean oil on the coated and uncoated fabrics; d) variation of liquid drops with time: 1) water, 2) soybean oil, and 3) hexadecane; e) dependency of CA and SA on the surface tension of liquids. All panels reproduced with permission.^[24] Copyright 2013, Wiley.

and then transfer efficient energy to HPU matrix, resulting in the melting, diffusion, and rearrangement of the soft segment of HPU to crack place. Therefore, the nanocomposites showed rapid and repeatable self-healing properties.

Anisotropic Ag nanoparticles (Ag NPs) are a kind of green dyes due to their well-tailored surface plasmon resonance. However, water soluble Ag NPs cannot be firmly deposited onto cotton fabrics especially for laundering process, which

leads to poor color fastness and restricts their practical applications.^[29] A highly superhydrophobic and water-repellent film on the cotton fabrics could solve this problem. Meanwhile, self-healing superhydrophobic properties are of great importance to the surfaces that suffer from chemical oxidation and physical damages.^[24] Moreover, as Ag NPs exhibit wide-spectrum bactericidal activity,^[30] it is possible to develop multifunctional materials based on anisotropic Ag NPs.

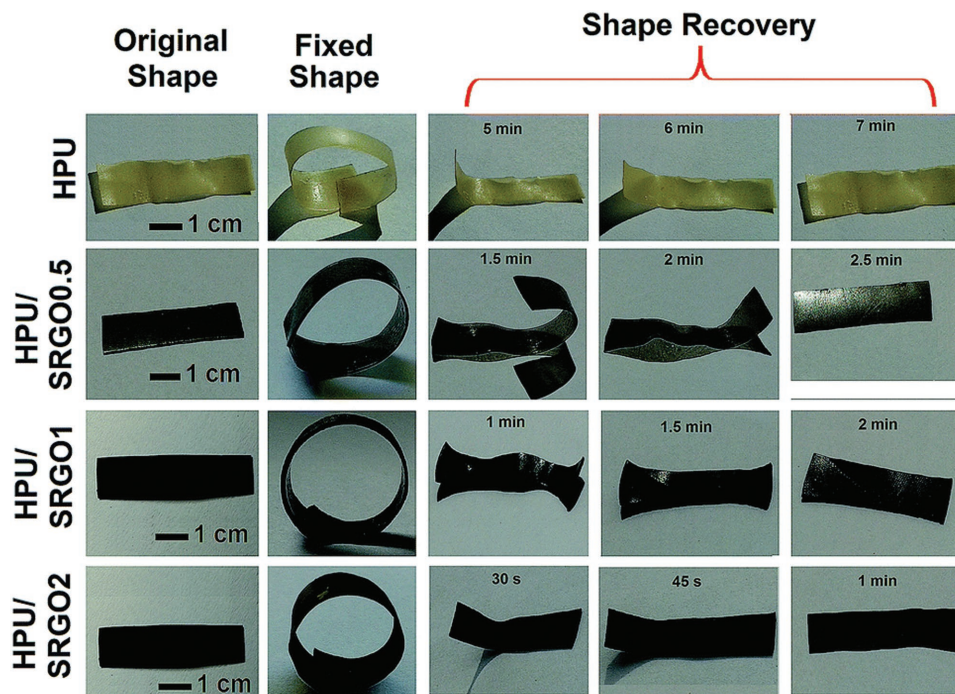


Figure 5. Shape-memory behaviors of HPU and HPU/SRGO under sunlight. Reproduced with permission.^[28] Copyright 2015, Royal Society of Chemistry.

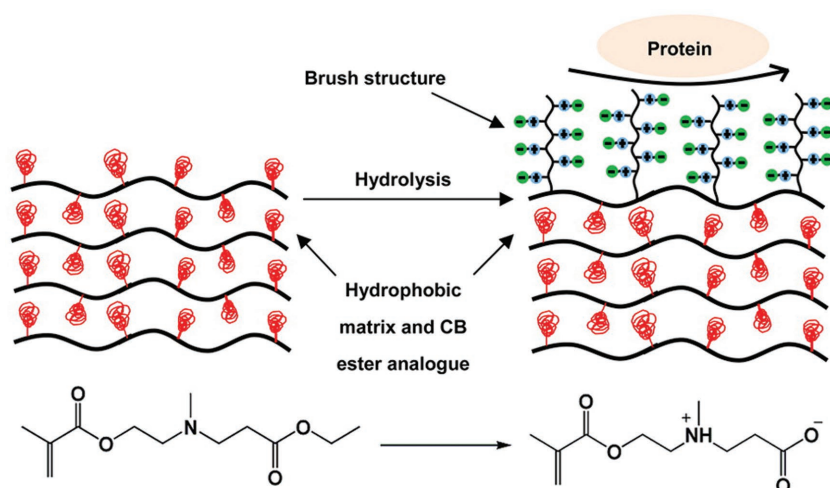


Figure 6. Schematic illustration of antifouling mechanism of PUs. Reproduced with permission.^[35] Copyright 2015, American Chemical Society.

In 2016, Sun and co-workers^[31] reported the fabrication of colored cotton fabrics with good color fastness, self-healing superhydrophobic and antibacterial properties. They used a convenient solution-dipping method to sequentially deposit branched poly(ethylenimine) (bPEI), Ag NPs, and fluorinated decyl polyhedral oligomeric silsesquioxane (F-POSS) on cotton fabrics. bPEI was used to facilitate the deposition of Ag NPs, while Ag NPs could then be deposited through electrostatic and hydrogen bonding interactions between the amino groups of the PEI and the carboxylate groups of the Ag NPs. F-POSS was further deposited to contribute high water-repellent ability. As F-POSS could also penetrate into the Ag NPs/PEI-coated cotton fabrics, when the coating was damaged by sunlight bleaching or physical rubbing, the embedded F-POSS could migrate to surface driven by the increase of the surface energy of the cotton fabric and formed a layer of F-POSS in a humid environment, maintaining the superhydrophobicity of cotton fabrics. These coated fabrics also exhibited satisfactory antibacterial properties. Such multifunctional cotton fabrics may have the potential to be used as protective clothing in moist and insanitary environments. However, consumption of the embedded F-POSS over damage/healing cycles slowed healing process down and it took 10 h to heal on the 16th damage/healing cycle.

In summary, all the materials introduced in this section exhibiting self-healing ability mainly rely on the migration or rearrangement of functional polymer chains. They may not restore their original structure, but their antifouling function keeps intact. These materials are usually used as coatings. However, complicated procedures are required to prepare most of these materials, and the coating thickness is only a few tens of nanometers, thus it is difficult to heal large damages.

2.2. Reaction-Based Materials

Similar to structural rearrangement-based antifouling and self-healing materials, reaction-based materials can also repair cracks or scratches via exposing the damaged area to solvents

or heat, while the difference is that reaction-based materials require external stimulus to trigger the reactions and regain the antifouling functions.

Zwitterionic polymers are expected to have better antifouling efficacies as they can bind water molecules more strongly than PEG.^[32] Chen and co-workers found that in basic solution, carboxybetaine ester could be hydrolyzed gradually to protein-resistant zwitterionic compounds. Based on this mechanism, they developed some hydrophobic zwitterionic precursor-based antifouling materials, such as silicone hydrogel^[33] and drug-loaded wound dressing.^[34] However, the nonfouling monomers in these materials would increase the undesired water content, which is bad for their mechanical properties and durability. To overcome this problem, in 2015, they reported a new kind of PU material containing poly(carboxybetaine) ester analog

side chains, which were hydrophobic when pH was below 7.4 (Figure 6).^[35] After hydrolysis, poly(carboxybetaine) ester-PU could resist protein adsorption, bacterial adhesion, and human umbilical vein endothelial cell attachment, and more importantly, the anti-biofouling property of poly(carboxybetaine) ester-PU was proved to be long-term stable and self-recoverable against scratches, as the antifouling properties in damaged sites could be healed through self-regeneration of the zwitterionic polymer brush coating on the surface by quick hydrolysis from hydrophobic matrix. Specially, the hydrolysis reaction can be carried out in alkaline solutions, physiological buffers (such as 0.15 M, pH 7.4 phosphate-buffered saline) or steam in an autoclave. Furthermore, this material showed good adhesiveness as paint on both polar and nonpolar substrates, making it an ideal candidate as marine and medical device coatings.

Similarly but excitingly, a new “hidden zwitterion” strategy is reported to achieve both high bulk mechanical strength and excellent surface nonfouling properties through a nonfouling polymeric elastomer based on zwitterionic polycarboxybetaine derivatives, without the need of surface coating.^[36] As shown in Figure 7, replacing the quaternary amines in conventional poly(carboxybetaine) with tertiary amines and protecting its carboxyl groups with esters convert the superhydrophilic and zwitterionic poly(carboxybetaine) into charge neutral and hydrophobic protonated tertiary carboxybetaine ester polymer (PTCBE). The resulting PTCBE also exhibited improved mechanical properties. Moreover, its surface can be hydrolyzed in a controlled manner under specific conditions to reveal the “hidden zwitterions” and generate a thin nonfouling layer, which is also the reason for healable ability. Choosing isobornyl as ester protecting group, the elastomer could effectively reduce attachment from fibrinogen, bovine aortic endothelial cells, platelets, and bacteria, and meanwhile showed excellent mechanical properties. Furthermore, a tube (or catheter) prototype was fabricated using the protonated tertiary carboxybetaine isobornyl elastomer, and nonfouling surface was achieved without sophisticated surface coating process. This coating-free nonfouling elastomer is a highly promising biomaterial for biomedical and engineering applications.

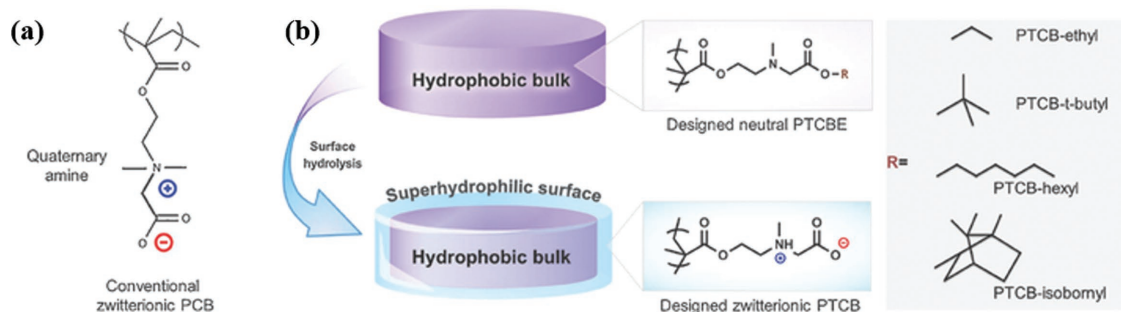


Figure 7. Schematics of nonfouling PTCBE elastomer: a) chemical structure of conventional zwitterionic poly(carboxybetaine); b) PTCBE elastomers with different types of carboxylate esters lead to a superhydrophilic surface upon surface hydrolysis. All panels reproduced with permission.^[36] Copyright 2017, Wiley.

N-halamines and cationic polymers are two kinds of antimicrobials, which have been demonstrated to be against several microorganisms. *N*-halamines could exhibit antimicrobial character under chlorinated state, mainly by releasing the Cl bonding with N, and the antimicrobial character can be recovered by repeated exposure to Cl source. However, the stability and antimicrobial efficacy of the coatings remain a significant challenge in the presence of organic matter.^[37] To solve this problem, Bastarrachea and Goddard prepared a healable, antimicrobial polymer coating composed of both *N*-halamine (styrene maleic anhydride copolymer, SMA) and cationic polymer (bPEI), and applied it to a maleic anhydride-functionalized polypropylene support (PP-g-MA).^[38] Briefly, bPEI and styrene maleic anhydride copolymer were sequentially applied on PP-g-MA, as shown in **Figure 8**, and the binding between the NH_2 of PEI and the maleic anhydride groups was evidenced by

IR. Then, the coating was formed after curing at 165 °C. This polymer coating retained efficacy against *E. coli* O157:H7 in organic matter. In the presence of up to 5000 ppm horse serum, both chlorinated and nonchlorinated coatings showed great antimicrobial effect. Additionally, after exposure to pH extremes (5% HCl or 4% NaOH), this material could heal by heating at 140 °C for 20 min, due to the reformation of acid- and base-hydrolyzed imide bonds, which was convinced by IR spectra.

Microencapsulation of healing agents is a useful strategy to develop self-healing materials. However, the solvent compatibility and thermal stability of MCs need to be improved in practical applications. Yang and co-workers^[39] reported a facial one-pot microencapsulation approach to efficiently prepare five reactive or nonreactive liquid cores in resorcinol-modified poly(urea-formaldehyde) (PUF) MCs with superior resistance to non- or less-polar solvents and thermal stability. Particularly,

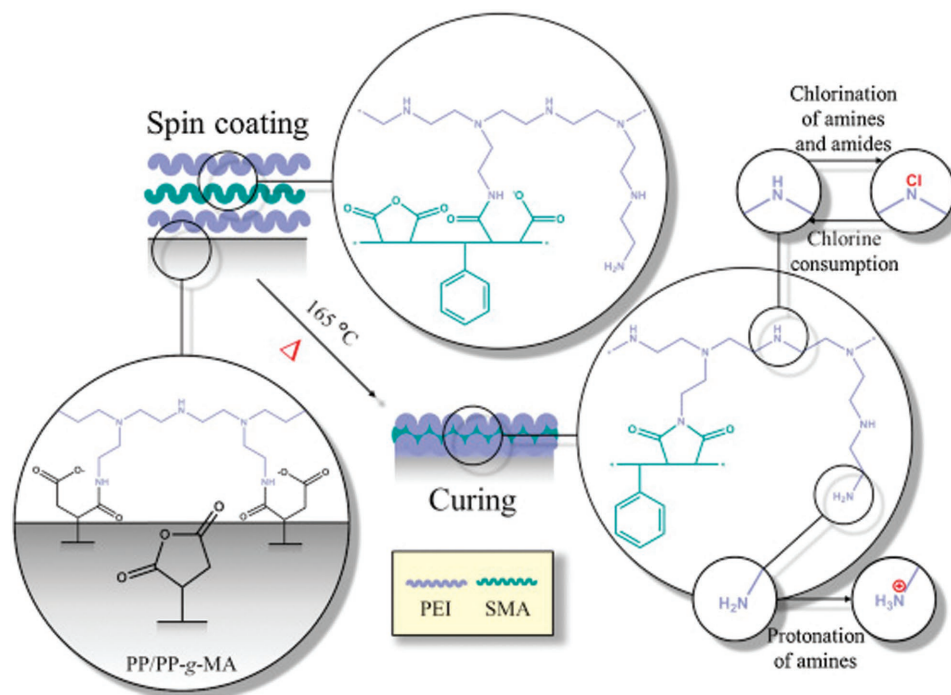


Figure 8. Preparation of the antimicrobial PP-PEI-SMA-PEI coating. Reproduced with permission.^[38] Copyright 2016, Elsevier.

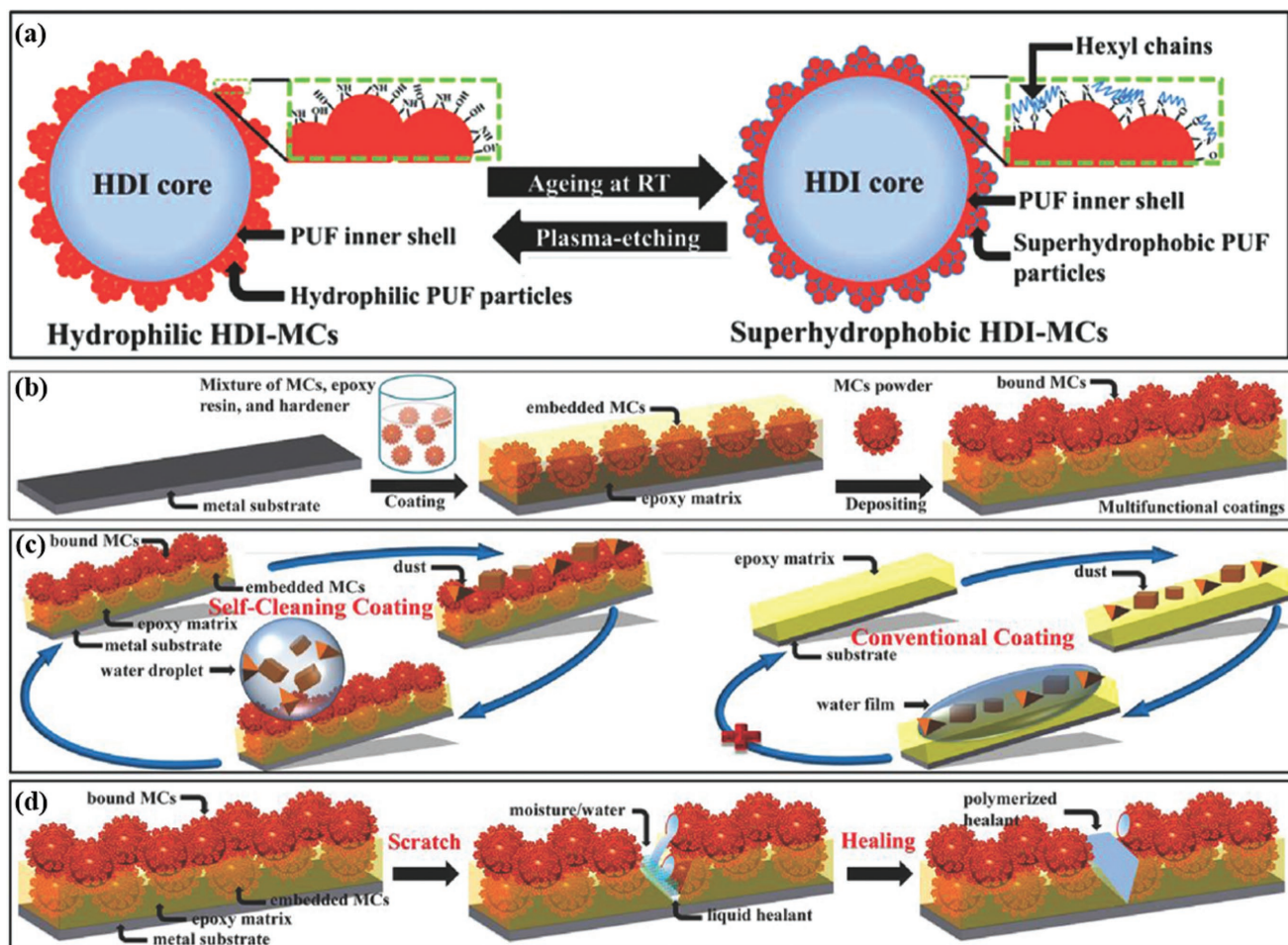


Figure 9. a) Schematic illustration of hydrophilic–superhydrophobic transition of HDI MCs; b) preparation of multifunctional coatings; c) the anti-fouling process; d) the self-healing process. All panels reproduced with permission.^[39] Copyright 2014, Wiley.

MCs containing hexamethylene diisocyanate (HDI) as a moisture sensitive healant showed superhydrophobicity. Self-healing and antifouling coatings could be fabricated by dispersing the HDI-filled MCs in a polymer matrix. Interestingly, the newly prepared HDI MCs were hydrophilic due to the abundant hydrophilic amino and hydroxyl groups on the surfaces of PUF particles during the formation of PUF MCs. HDI could react with water or moisture to generate unstable intermediate carbamic acids that decomposed to amines, and the generated amines in turn reacted with other HDI molecules to form polyureas. As a result, the surface gradually changed to superhydrophobic (Figure 9a). When dispersed in polymer matrix, the self-healing ability was realized by the release of HDI and then reaction with water to form polyurea to fill the cracks (Figure 9b–d).

In summary, all the materials mentioned above obtain their healing ability with the aid of external stimuli, such as water and heat. The reaction mainly occurs at the surface, while the bulk is inert. Like structural rearrangement-based materials, the reaction-based materials can ensure the antifouling functions, rather than structural integrity. Moreover, the thickness of these materials is not a limitation any more, which will extend the application of such kind of materials.

2.3. Ionic Bond–Based Materials

The ionic bond–based materials utilize ionic crosslinkings between anionic polymers and cationic species such as metal ions, small molecules, or cationic polymers. The damaged ionic crosslinking networks can be immediately restored in response to external stimuli (mainly water) at an ambient condition. The healing mechanism of polyelectrolyte films and zwitterionic polymer–based films is mainly based on ionic interactions.

PEG- or zwitterionic polymer–based antifouling films fabricated by surface-initiated polymerization or covalent coupling methods on solid substrates are usually several tens of nanometers thick. And, these thin films cannot stand severe mechanical damages of several tens of micrometers wide. Recently, layer-by-layer (LbL) assembly, achieved by alternate deposition of species, has been used to fabricate self-healing films.^[40] The properties of LbL-assembled film could be finely tuned by changing the assembly parameters during film deposition, such as temperature, ionic strength, concentration, pH of the dipping solutions, and the number of layers deposited. Endowing LbL-assembled polyelectrolyte multilayer films in micrometer thick with anti-biofouling properties, several anti-biofouling and healable films were successfully produced.

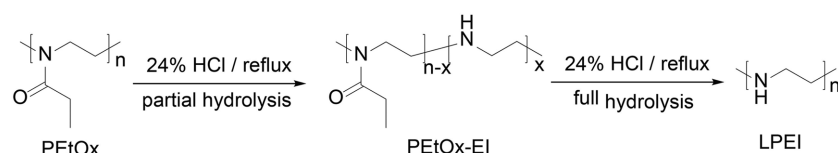


Figure 10. Synthetic route of the partially and fully hydrolyzed PEOx.

For example, Sun and co-workers reported a series of LbL-assembled healable antifouling films based on high diffusion of polyelectrolytes in the presence of water and reversible electrostatic interactions. In 2014, they fabricated an optically transparent antibacterial film which could heal scratches and restore transparency by LbL assembly of bPEI/poly (acrylic acid) (PAA) films and then postdiffusion of antibacterial agent triclosan-encapsulated cetyltrimethylammonium bromide micelles.^[41] The sustained release of triclosan molecules can inhibit the growth of both gram-positive and gram-negative bacteria. Particularly, scratches on the films could be completely repaired in water, as bPEI and PAA could flow and recombine in water through electrostatic and hydrogen bonding interactions. The optical transmission of the films could also be restored once the films were dry. All these features made this film a potential candidate as transparent protective film with antibacterial properties.

In 2015, another antifouling healable film was fabricated by LbL assembly of PEGylated bPEI (i.e., PEG was conjugated onto bPEI backbone) and hyaluronic acid (HA) for 20 deposition cycles.^[42] The bPEI-PEG/HA films were then crosslinked by PEG bis(carboxymethyl) ether as they are unstable in normal saline. This crosslinked film could effectively prevent protein adsorption and cell attachment, the softness of crosslinked bPEI-PEG/HA films in normal saline helped to resist cell adhesion and the PEG-rich surface further enhanced the resistance effect. In addition, the restoration of antifouling property of the crosslinked bPEI-PEG/HA films was demonstrated, SEM images showed that the cut with a width of $\approx 100\ \mu\text{m}$ on the film disappeared after immersing it in saline for 5 min, as the mobility of bPEI-PEG and HA polyelectrolytes and the reversible electrostatic and hydrogen bonding interactions between them drove the reformation of crosslinked bPEI-PEG/HA

antifouling films and thus rapidly healed the damages in physiological conditions.

PEG-containing antifouling films usually suffer from auto-oxidation and formation of aldehydes and ethers in the presence of oxygen and transition metal ions, leading to the loss of antifouling properties.^[43]

Poly(2-ethyl-2-oxazoline) (PEOx) and its derivatives exhibit comparable antifouling ability to PEG, while they are more stable than PEG in oxidative environments.^[44] As shown in **Figure 10**, PEOx could be gradually hydrolyzed in HCl solution at $100\ ^\circ\text{C}$, with amide groups converting into secondary amine groups. Through LbL assembly of partially hydrolyzed PEOx and PAA followed by thermal crosslinking, another healable antifouling film was fabricated.^[45] Research showed that partially hydrolyzed PEOx with hydrolysis degree of 7% (PEOx-EI-7%) could obtain the thickest film under the same deposition cycles. PEOx-EI-7% with high surface density could bind water to form a hydration layer, and the film exhibited highly effective antifouling abilities that prevented adhesion of Gram-negative *E. coli* and Gram-positive *B. subtilis* bacteria. This film was able to heal cuts of a few tens of micrometers wide in the presence of water due to the high mobility of the polymer chains and their hydrogen-bonding interactions (**Figure 11**). When immersing the damaged (PAA/PEOx-EI-7%)**n* films in water, the formation of hydrogen bonds between water and PEOx-EI-7% and PAA partially broke the hydrogen-bonding interactions between PEOx-EI-7% and PAA to make the films soft and flowable. The high mobility of the films facilitated the immediate contact of fractured surfaces, and then the reformation of the hydrogen-bonding interactions between PAA and PEOx-EI-7% in the fractured surfaces helped to heal the film completely.

Different from the LbL assemble approach but with a similar healing mechanism, Kundu and co-workers prepared a self-healable polyelectrolyte film via crosslinking the cationic chitosan with anionically modified bacterial cellulose (BC).^[46] This polyelectrolyte film showed self-healing activity due to ionic interactions between cationic NH_3^+ and anionic $-\text{COO}^-$ in the film. When physiological buffer solution was sprinkled over the composite film, the polymeric segments (cationic

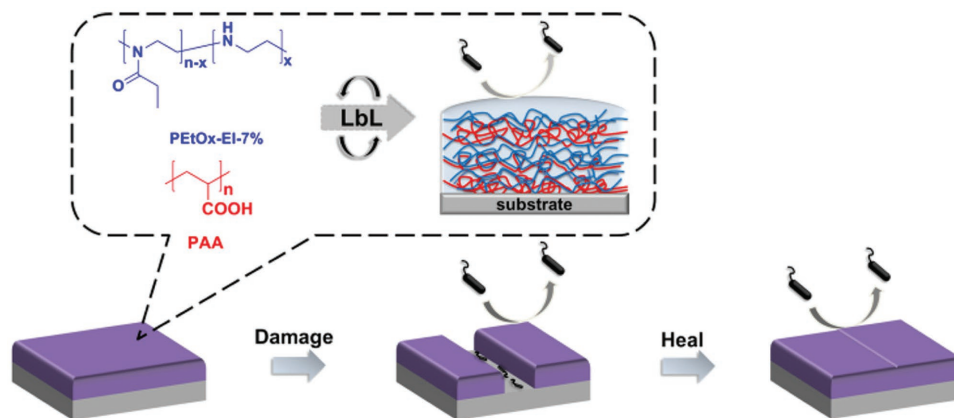


Figure 11. Schematic of the healable antifouling films composed of partially hydrolyzed PEOx and PAA. Reproduced with permission.^[45] Copyright 2017, American Chemical Society.

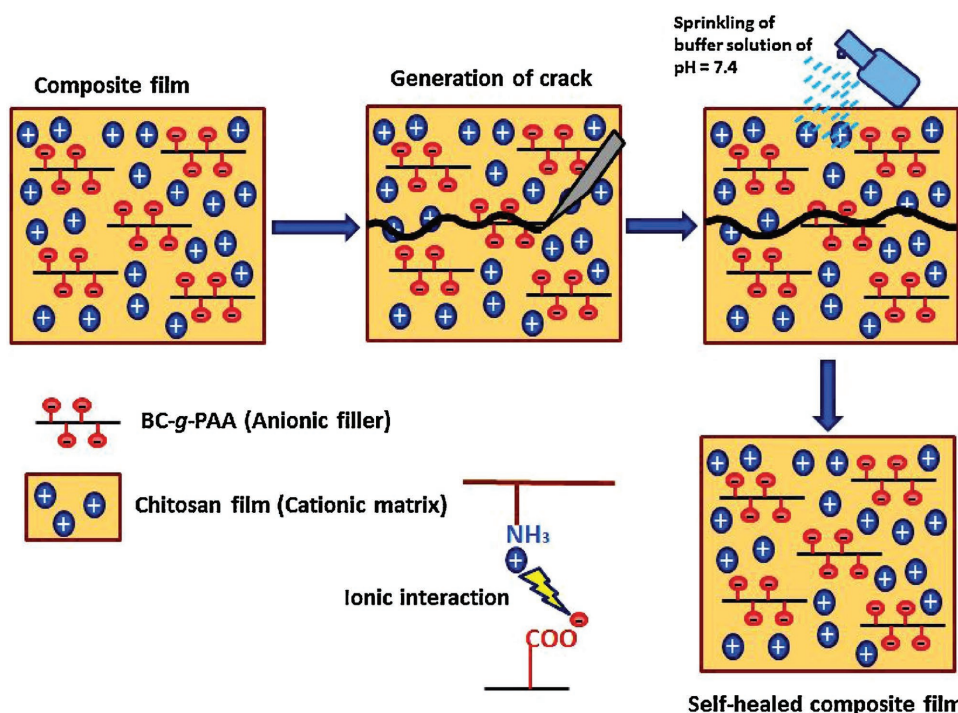


Figure 12. Mechanism of self-healing assisted by anionic filler (BC-g-PAA) in cationic chitosan matrix. Reproduced with permission.^[46] Copyright 2017, Elsevier.

chitosan) were tending to diffuse in the bulk of the system and form an ionic interaction with the anionic filler (**Figure 12**). Furthermore, this film could encapsulate and deliver curcumin which had antimicrobial and wound healing capability.

Coating surfaces with zwitterionic materials is also an effective way to reduce or eliminate biofouling.^[47] Zuilhof

and co-workers^[48] prepared zwitterionic polymer network coatings by copolymerizing 2-[(methacryloyloxy)ethyl] dimethyl-(3-sulfopropyl)-ammonium hydroxide and poly(ethylene glycol) dimethacrylate in water (**Figure 13**). The great antifouling property was attributed to 2-[(methacryloyloxy)ethyl] dimethyl-(3-sulfopropyl)-ammonium hydroxide. When mechanically

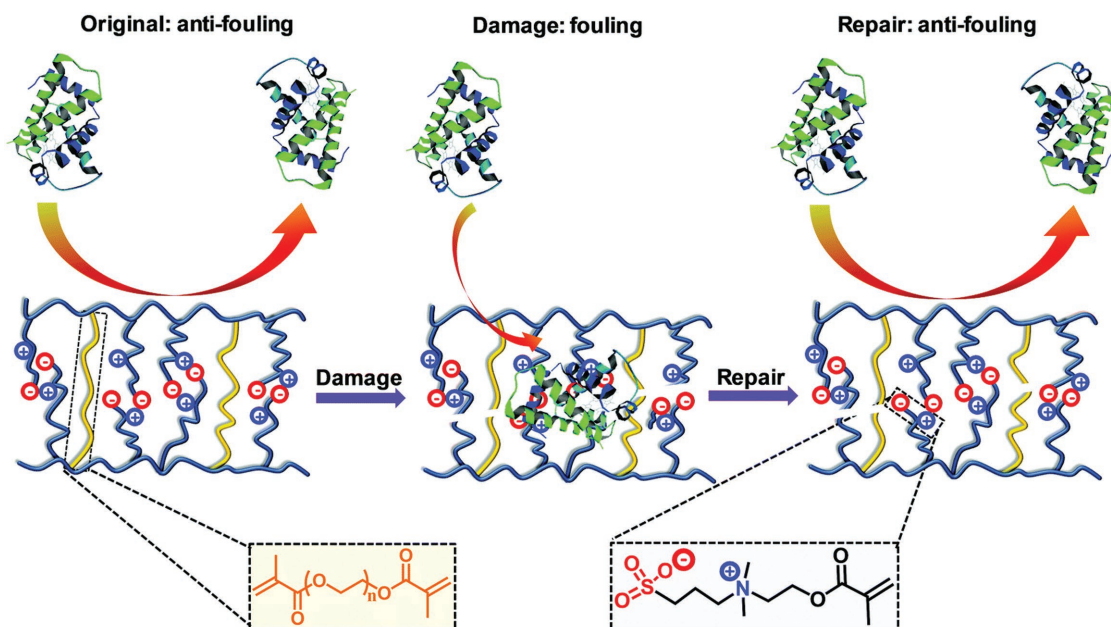


Figure 13. Schematic illustration of the monomer molecules, self-healing process, and antifouling properties of zwitterionic polymer networks. Reproduced with permission.^[48] Copyright 2017, Royal Society of Chemistry.

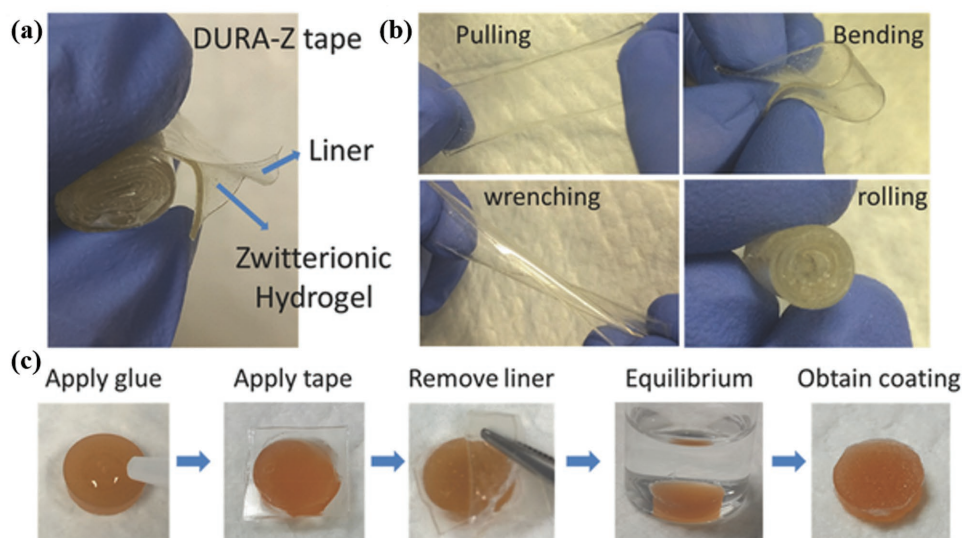


Figure 14. a) Structure of DURA-Z tape; b) DURA-Z tape under pulling, bending, wrenching, and rolling; c) fabrication of DURA-Z coating on PU substrate. All panels reproduced with permission.^[49] Copyright 2017, Wiley.

damaged, the coatings could be recovered by immersing in water within 1 min, due to the reformation of the broken ionic bonds between the protonated ammonium and sulfonic group in the presence of water. Especially, this self-healing property could be realized in nanoscale thickness, and the mechanical as well as anti-biofouling properties could also be repaired.

Nonetheless, similar to common hydrophilic coatings, zwitterionic materials are liable to dissolve in the aqueous environment, resulting in poor durability.^[35] The vulnerability to mechanical damages also limits their practical applications. To solve these problems, a durable and ultrarobust antifouling zwitterionic (DURA-Z) coating by using a double-sided DURA-Z tape and superglue was reported lately.^[49] The DURA-Z tape was prepared by in situ growth of a thin layer of poly[3-((3-acrylamidopropyl) dimethylammonio) propanoate]] hydrogel^[49] on a commercially available polypropylene liner (Figure 14). The commercial superglue based on ethyl cyanoacrylate was used to immobilize superhydrophilic DURA-Z through interpenetration mechanism. This coating showed great stability and durability in a series of long-term challenges in aqueous environments, such as 10 dry/wet cycles, 90 days of immersion in water, 50 days of buffer shearing, and 30 days of water flushing. Additionally, the damaged DURA-Z coating layer was able to heal together in water as formation of zwitterionic pairs via electrostatic interactions between positively and negatively charged zwitterionic moieties can reconnect the cut pieces.^[50] Excitingly, it remained almost “zero” microbe adhesion in the presence of more than 10^9 cells mL^{-1} culture medium for 30 days. All the above advantages made this simple and inexpensive material a promising candidate as antifouling coating in practical use, such as anti-biofilm coatings, medical device coatings, and marine coatings.

Ionic bond-based healable and anti-biofouling materials, mainly films, exhibit self-healing in the presence of water, due to the high chain mobility and ionic interactions (some also with the aid of hydrogen bonding). These films have tunable structure properties, and the healing can be realized from

several tens of nanometers to hundreds of micrometers. The antifouling properties come from either zwitterionic polymers themselves, or covalent antifouling polymers and noncovalent antifoulants in the polyelectrolyte films.

2.4. Anti-Biofouling and Self-Healing Hydrogels

Hydrogels, especially injectable hydrogels, have been considered as attractive candidates for various medical applications such as drug delivery, tissue engineering, wound filler, and implant coating, because of their biocompatibility, high water content, and similar structure to organisms.^[51] Though great development has been made for self-healing hydrogels, the practical applications still face a lot of problems. For example, proteins or microorganisms might adhere and form biofouling films on hydrogels, leading to immune response or inflammation during in vivo applications, especially those associating with impaired wound healing.^[52] Hence, hydrogels with anti-biofouling properties attract great attentions in biomedical researches.^[53] Furthermore, the implanted hydrogels are subjected to constant mechanical forces from daily body movements which would cause gel deformations or even damages. Thus, the development of effective antibacterial and anti-inflammatory hydrogels with structural and functional integrity, i.e., self-healing ability has become an urgent task.

Generally, there are two strategies to prepare self-healing hydrogels. One is based on noncovalent or physical crosslinking, such as hydrophobic bonding, hydrogen bonding, ionic bonding, and host–guest interactions. The other is based on the dynamic covalent bonding (chemical crosslinking), such as imine bonds, acylhydrazone bonds, boronate ester bonds, and disulfide bonds. In both the systems, the continuous and dynamic equilibrium between dissociation and recombination of bonds confers self-healing ability on the hydrogels.^[11a,54]

As hydrogen bonding is relatively weak, the most commonly used self-healing hydrogel preparing approaches utilize multiple

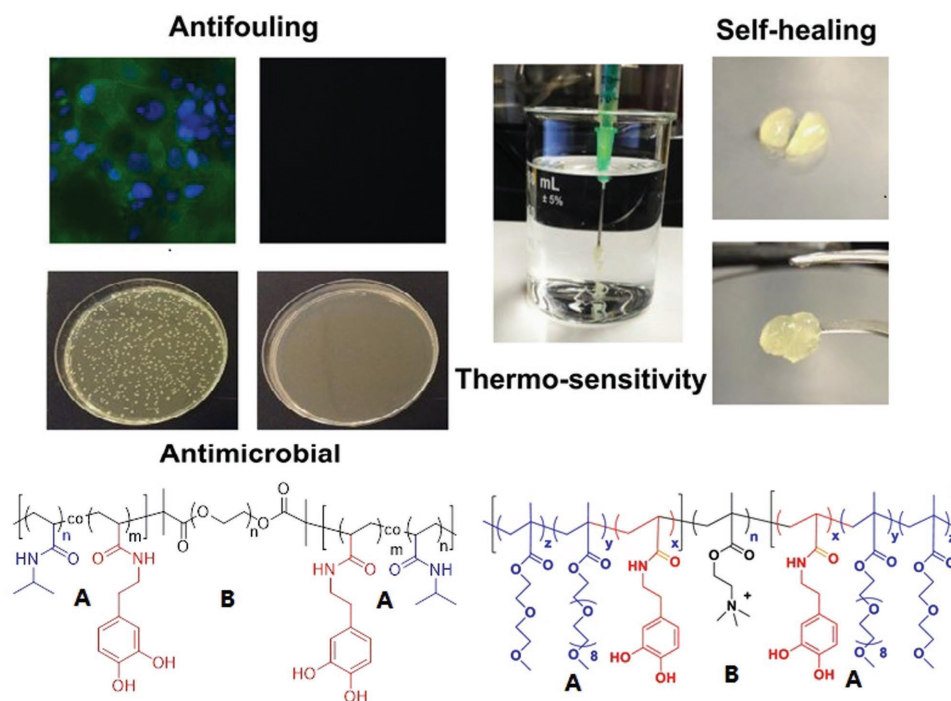


Figure 15. Two injectable self-healing hydrogels based on ABA triblock copolymers with antimicrobial and antifouling properties. Reproduced with permission.^[56] Copyright 2017, American Chemical Society.

or cooperative hydrogen bonding interactions to assemble integrate structures. Recently, Zeng and co-workers reported two injectable self-healing hydrogels with antimicrobial and antifouling properties through self-assembly of ABA triblock copolymers in metal-free aqueous environment.^[55,56] Both poly(*N*-isopropylacrylamide) (PNIPAM) (A block) and hydrophilic PEO (B block) were functionalized by mussel-inspired catechol. Due to the mussel-inspired hydrogen bonding and aromatic interactions mediated by catechol, as well as thermosensitivity, these hydrogels demonstrated a fully reversible sol–gel transition behavior at 4 °C (or 0 °C) and 37 °C, that could heal the damages autonomously.^[55] In addition, poly[[2-(methacryloyloxy)-ethyl] trimethylammonium iodide] and PEO imparted the excellent antifouling performance to the hydrogels, as shown in **Figure 15**. More importantly, compared with catechol–metal complex–based mussel-inspired self-healing hydrogels, these materials showed less cytotoxicity. All these advantages endowed the hydrogels with great potential in bioengineering applications.

N-acryloyl glycineamide (NAGA) has been demonstrated as physical crosslink due to the stable multiple hydrogen bonding formed by the dual amide in side chain of NAGA.^[57] Recently, a supramolecular polymer hydrogel was synthesized by the copolymerization of NAGA and 1-vinyl-1,2,4-triazole (VTZ) without chemical crosslinker (**Figure 16**).^[58] Both experimental and Gaussian calculation results indicated that the introduction of VTZ changed the hydrogen bond interactions and endowed the hydrogel with self-repairing ability, thermoplasticity, and reprocessing ability for 3D printing. In addition, the triazole ring nucleus of VTZ made NAGA–VTZ hydrogels antimicrobial and anti-inflammatory. Heating-triggered healing ability of NAGA–VTZ hydrogel was also observed, which was attributed to the reformation of hydrogen bonds at the broken interfaces.

Incorporating multiple networks of physical crosslinks into a hydrogel can improve both self-healing and mechanical properties. A dual physically crosslinked hydrogel has been developed by doping Ln^{3+} (lanthanide ions, i.e., Eu^{3+} , Tb^{3+}) into the Na-alginate/polyvinyl alcohol (PVA) hydrogel.^[59] In this system, hydrogen bonds were formed among PVA polymers, and coordination bonds were formed between Na-alginate networks and Ln^{3+} (**Figure 17**). Ln^{3+} has three functions, including endowing antibacterial hydrogels with low toxicity, photoluminescent property, and acting as crosslinker for alginate. The hydrogel exhibited high mechanical strength and healability, as the coordination bonds formed between trivalent Ln^{3+} ions

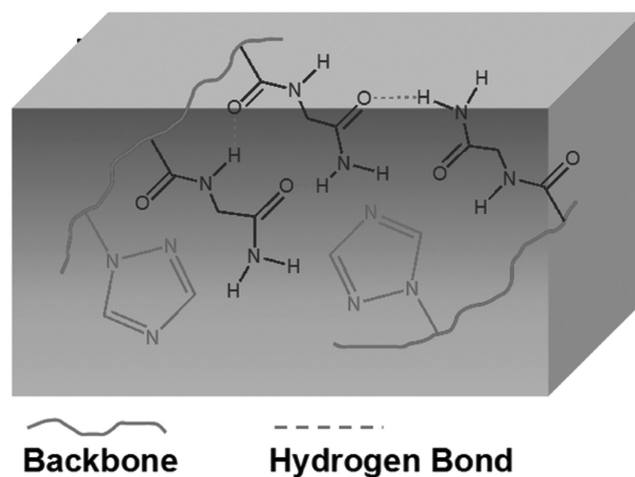


Figure 16. Schematic network structure of PNAGA–PVTZ hydrogel. Reproduced with permission.^[58] Copyright 2017, Wiley.

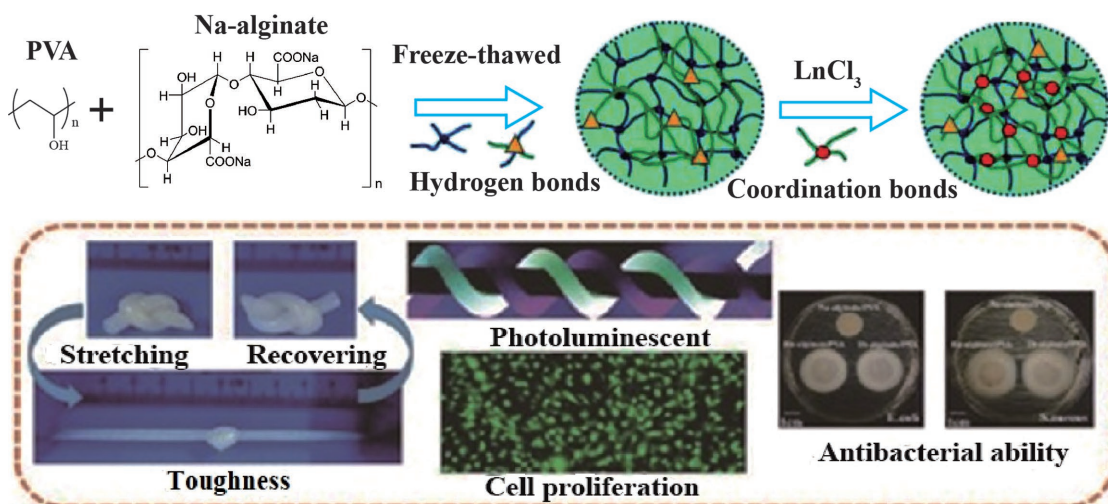


Figure 17. Schematic preparation of tough photoluminescent hydrogels and photographs of the knotted, stretched knotted, bending, and compressed Eu-alginate/PVA hydrogels with photoluminescence and antibacterial activities. Reproduced with permission.^[59] Copyright 2017, Wiley-VCH.

and the carboxyl ligands of Na-alginate could serve as physical sacrificial bonds for energy dissipation, while the networks crosslinked by hydrogen bonds remained intact. Furthermore, the emission colors of the hydrogels can be tuned by doping different lanthanide elements.

Double network is another approach to improve the mechanical properties of hydrogels.^[60] Recently, an antibacterial double network hydrogel was developed by the UV-initiated copolymerization of PAA-grafted quaternized cellulose and PVA.^[61] In this system, cellulose functioned as antibacterial agent, *N,N'*-methylene diacrylamide acted as chemical crosslinker, and PVA functioned as a physical crosslinker (**Figure 18**). The self-healing property was attributed to ionic bonding between carboxylic groups and ferric ions, as well as hydrogen bonding in PVA.

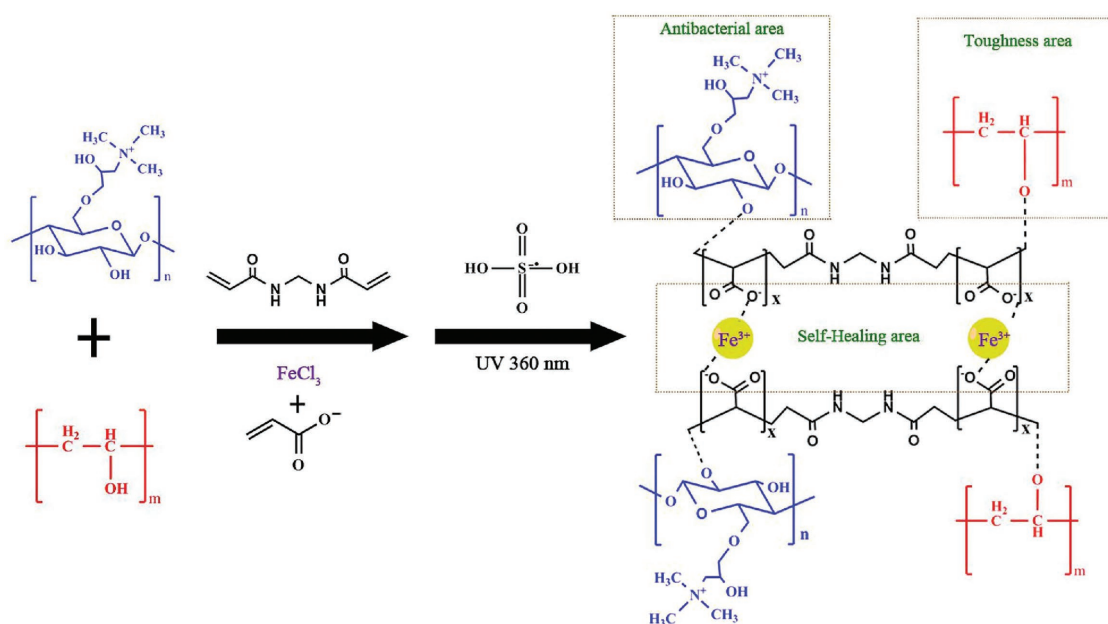
Imine bonds, also called Schiff base, are formed by amine groups and aldehyde groups.^[62] Recently, Guo and co-workers reported a series of biocompatible hydrogels with simultaneous injectability, conductivity, and antibacterial effect.^[63,64] Among them, two self-healing hydrogels, chitosan-graft-aniline tetramer/dibenzaldehyde-terminated PEG (CS-AT/PEG-DA) hydrogel and quaternized chitosan-g-polyaniline/benzaldehyde group-functionalized PEG-co-poly(glycerol sebacate) (QCSP/PEGs-FA) hydrogel were designed based on dynamic imine bonds that formed between the modified chitosan and benzaldehyde functionalized PEG (**Figure 19**).^[63,64] Among the building blocks, chitosan has antibacterial activity that can be improved by quaternized modification. The introduction of biocompatible and conductive aniline tetramer or polyaniline also brought the hydrogels a quite similar conductivity to nature tissues. Importantly, these hydrogels have been demonstrated to be potentially applied in wound healing and cardiac tissue repair.

Recently, a double network hydrogel was fabricated using an enzyme-induced route.^[65] In this system, radical polymerization of 1-vinyl-2-pyrrolidinone (NVP) and *N*-methylol acrylamide (NMA) in the presence of ϵ -poly-L-lysine (EPL) produced EPL-g-poly(NVP-co-NMA) hydrogels (**Figure 20**). The introduction of plasma amine oxidase in the system efficiently enhanced the

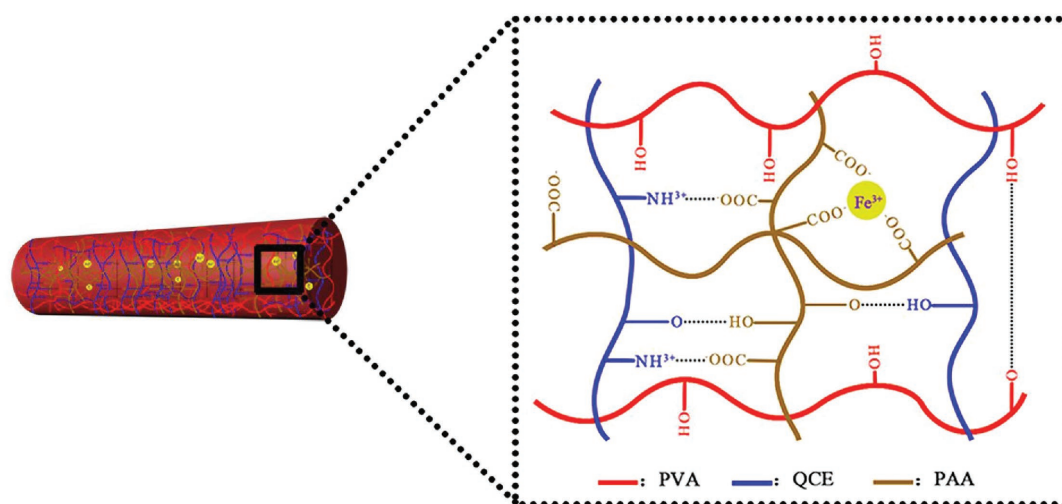
healing rate of the hydrogels. With the catalyzation of plasma amine oxidase, the primary amines of EPL were oxidized to aldehyde and then reacted with the remaining amino groups of EPL to generate Schiff base structure as secondary crosslinkings to enhance the formation of network. This dynamic Schiff base crosslinking structure highly enhanced self-healing efficiency (95%) by cooperating with the intrinsic hydrogen bonds. This hydrogel possessing broad-spectrum antimicrobial activity as EPL was an inherently antibacterial polypeptide biomaterial. In vivo wound healing tests of rat skin demonstrated the hydrogels' promising candidates as wound dressings.

Generally, self-healable materials require high chain mobility and appropriate amount of reversible bonds to support the recombination after break, which is realized by lower T_g or adding solvent. However, the poor mechanical properties limit their applications. Recently, a multifunctional self-healing and antibacterial coating with high mechanical strength was reported based on the dynamic interaction of vinyllogous urethane in multiamine hyperbranched polysiloxane (HPSi) and sulfobetaine groups in linear polyacrylate resin (LP) (**Figure 21**).^[66] Research showed that the amount of HPSi significantly affected the mechanical, thermal, and self-healing properties; the antibacterial rate was higher than 95% at the optimal HPSi amount of 6.3 wt%.

Disulfide bonds are thiol-disulfide exchange reaction in neutral or alkaline environment. A self-healing hydrogel was prepared on stainless steel surface, through thiol-ene photopolymerization (**Figure 22**).^[67] Disulfide bonds containing bis(2-methacryloyl)-oxyethyl disulfide (BMOD) acted as crosslinker and facilitated healing process. Poly(ethylene glycol) methyl ether methacrylate (PEGMA), *N*-hydroxyethyl acrylamide (HEAA), and 2-(methacryloyloxy)-ethyl trimethylammonium chloride (META) mainly imparted antifouling and antibacterial properties to the hydrogels. Self-healing of $\sim 10\ \mu\text{m}$ width cuts on thin hydrogel coatings was observed by AFM, results showed that the scratch could be completely healed in 48 h with small amount of water dropped onto the interfaces. The healing mechanism was ascribed to rebonding of the disulfide bonds



Synthesis of cellulose-based hydrogels



3D cellulose-based hydrogel

Figure 18. Schematic preparation of cellulose-based hydrogels. Reproduced with permission.^[61] Copyright 2017, Elsevier.

through disulfide exchange reaction at the cut interface. However, the disulfide bonding would lose dynamic properties in the long run, due to the oxidization of thiol when exposed to air.

Nanocomposite hydrogel is another way to develop self-healing and mechanically robust hydrogels. Generally, nanocomposite hydrogels consist of inorganic nanocomponents and soft, deformable, and elastic 3D hydrogel networks. The inorganic nanocomponents are uniformly dispersed into the polymer network and act as crosslinkers, sometimes they can even introduce special functions such as antifouling ability. For example, an antifouling self-healing nanocomposite hydrogel prepared by UV-initiated copolymerization of 2-hydroxyethyl methacrylate (HEMA) and zwitterionic nonfouling carboxybetaine methacrylamide (CBMAA-3) in the presence of clay

nanoparticles in water was reported in 2013.^[68] Zwitterionic CBMAA-3 endowed the hydrogel with not only antifouling property, but also self-healing capability. The mechanical properties could be adjusted by changing the concentration of zwitterionic comonomers and nanoparticles. The hydrogels showed good mechanical properties and elongations at break as high as 1800% even with higher molar contents of CBMAA-3. Such excellent properties made the hydrogels biocompatible, chemically stable, while low water content of poly(HEMA) and accessible to the applications in nature tissues. Another two nanocomposite hydrogels were prepared by the chelation between thiol/acetyl thioester groups and Ag NPs.^[69] The Ag NPs played two roles in the hydrogels, one as crosslinkers to impart healing ability and the other as antimicrobials to

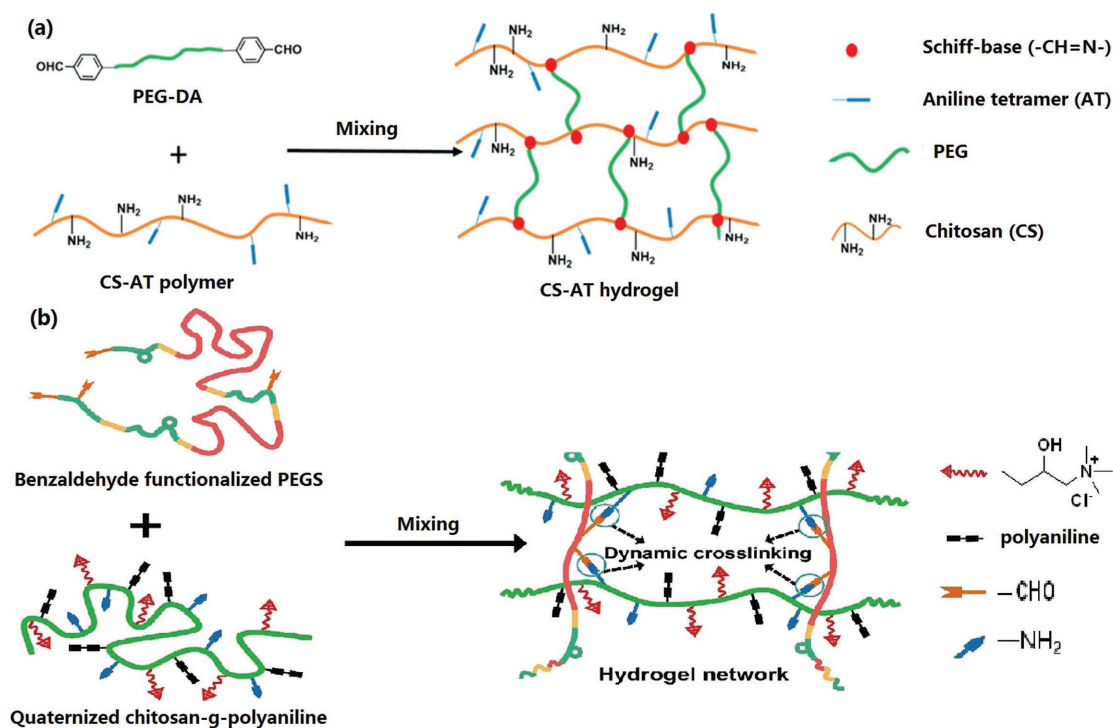


Figure 19. Schematic representation of dynamic imine bond-based hydrogels reported by Ma and co-workers: a) using aniline tetramer-modified chitosan and dibenzaldehyde-terminated PEG. Reproduced with permission.^[63] Copyright 2016, American Chemical Society; b) using polyaniline-functionalized quaternized chitosan and benzaldehyde-functionalized PEG-co-poly(glycerol sebacate). Reproduced with permission.^[64] Copyright 2017, Elsevier.

resist the algal adhesion. Due to the strong chelation between the thiol/acetyl thioester groups and Ag NPs, Ag NP leakage could be effectively overcome (Figure 23). The amount of Ag NPs embedded in the nanocomposite hydrogels significantly affected the mechanical properties. When the molar ratio of the Ag to the sulfur element was 0.9, the hydrogels showed the highest mechanical properties. The nanocomposite hydrogels containing thiol groups could self-heal more efficiently than the hydrogels with acetyl thioester groups.

In summary, self-healing hydrogels have many advantages compared to conventional hydrogels, such as longer lifetime and enhanced mechanical properties, by avoiding (or decreasing) of irreversible damage. Due to the lack of an adopted standard to demonstrate and quantify self-healing, it is difficult to make a comparison between each study. Nevertheless, developing a multifunctional hydrogel with appropriate mechanical strength, anti-biofouling, and self-healing ability are still the main targets in this research field.

2.5. Other Antifouling Healable Materials

Incorporation of healing agent containing microcapsules into matrix could also display healing efficacy.^[70] Due to the esthetics and direct-filling capability, the composites are broadly used in dental practice as filling materials. Generally, dental composites are composed of a resin matrix and inorganic filler particles as reinforce agent. Nonetheless, fracture and secondary caries are the main reasons for dental restoration failure.^[71] Caries are mainly caused by acids stemming from biofilm formed by

colonized bacteria.^[72] To solve these problems, a dental composite was developed in 2015 by incorporation of microcapsules with self-healing, antibacterial, and remineralization capabilities.^[73] Briefly, the microcapsules were synthesized with PUF shells containing triethylene glycol dimethacrylate and *N,N*-dihydroxyethyl-*p*-toluidine as healing liquid; in the resin matrix, dimethylaminohexadecyl methacrylate possessed antibacterial function, amorphous calcium phosphate nanoparticles were used for calcium and phosphate ion release and remineralization, and glass fillers were used to improve mechanical strength. Importantly, the mechanical properties of the composite were not affected by addition of even 7.5% of microcapsules, and 65–81% recovery in the virgin fracture toughness could be achieved when damages happened. Furthermore, this composite exhibited strong antibacterial effect against the formation of biofilms, inhibiting lactic acid production, and then inhibiting secondary caries. Nonetheless, whether the dimethylaminohexadecyl methacrylate would exhibit a long-term antibacterial effect needed further investigation.

Silicone polymer especially poly(dimethylsiloxane) (PDMS) elastomer-based coatings, can effectively inhibit or release biofouling due to their low surface energy and elastic modulus.^[74] However, their antifouling performance is limited either under static conditions, or due to poor mechanism properties and adhesion to substrates.^[75] Steric repulsion and the hydration layer mechanism have been developed to obtain biocompatible antifouling coatings of PDMS.^[76] Recently, a self-repairing coating consisting of PDMS-based polyurea and antifoulant (4,5-dichloro-2-*n*-octyl-4-isothiazolin-3-one) (DCOIT) was reported.^[77] The mechanical properties of PDMS-based

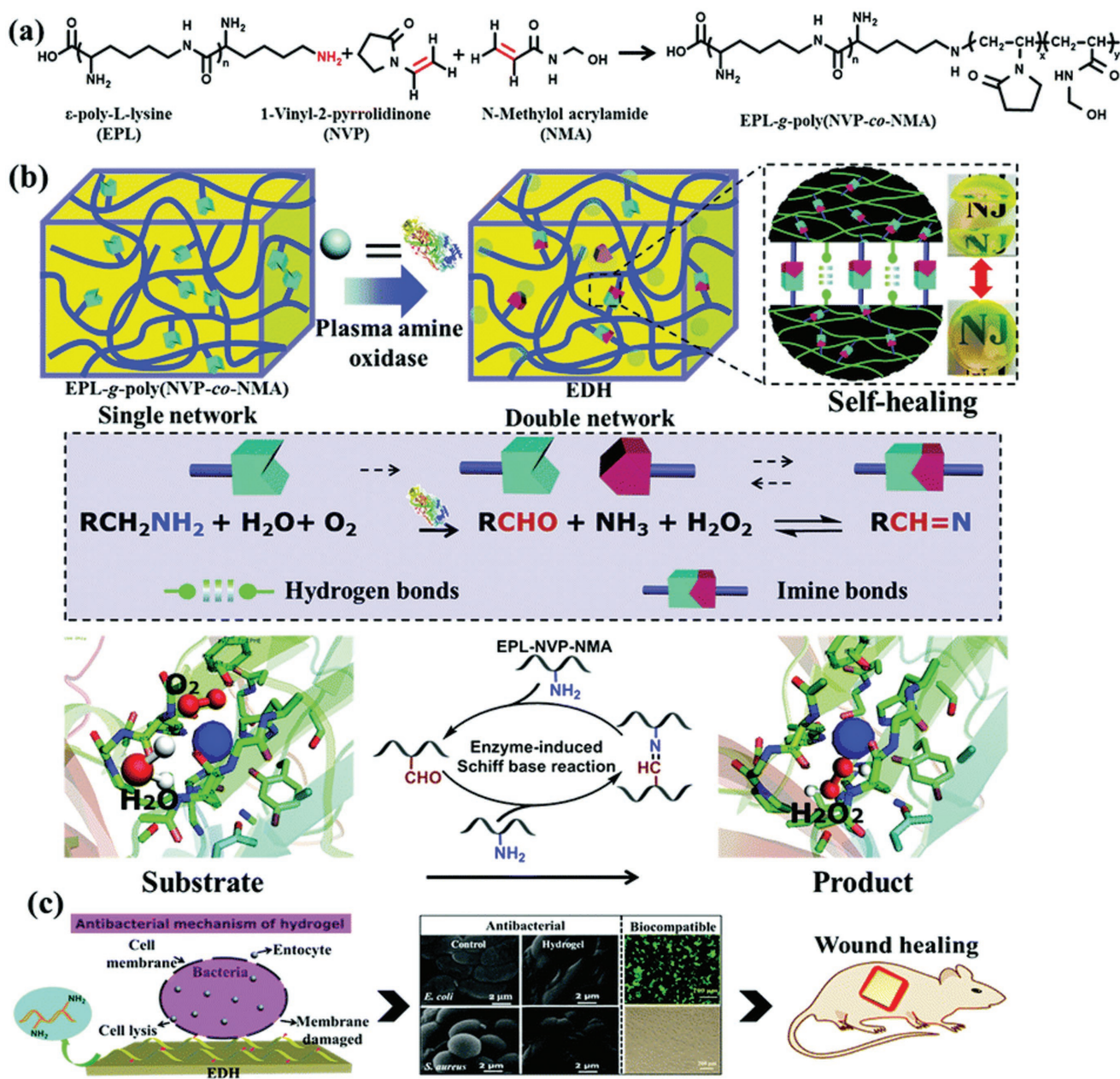


Figure 20. a) Synthesis of EPL-g-poly(NVP-co-NMA) hydrogels; b) schematic preparation of the enzyme-assisted dual-network self-healing hydrogels; c) antibacterial applications. All panels reproduced with permission.^[65] Copyright 2017, Royal Society of Chemistry.

polyurea was not affected by the addition of DCOIT. Due to the dynamic hydrogen bonding formed by polyurea, this coating could recover its mechanical properties in 48 h after damage at room temperature and within a shortened time at a higher temperature. Moreover, the release rate of DCOIT was almost constant and could be regulated by its concentration. Marine field tests demonstrated that the coating had excellent antifouling/fouling release performance for more than six months, indicating their potentials in marine anti-biofouling.

Generally, there are three kinds of anti-biofouling materials classified based on their functions: i) fouling-resist surfaces, which prevent the initial attachment of foulants, for example,

PEG- and zwitterionic polymer-based coatings; ii) microorganism-killing surfaces, which kill attached biofoulants by chemical agents; iii) fouling-release surfaces, which have minimal adhesion between foulants and accelerate the release of foulants by external force. All the anti-biofouling and healable materials mentioned above belong to fouling-resistant and microorganism-killing surfaces. Recently, dual-functional anti-biofouling surfaces which integrate at least two of the above functions have also been fully developed and demonstrated to prolong the working life.^[78] Meanwhile, the unique healing of antifouling properties can be found in such kind of materials. Kang and co-workers reported a dextran- and chitosan-based

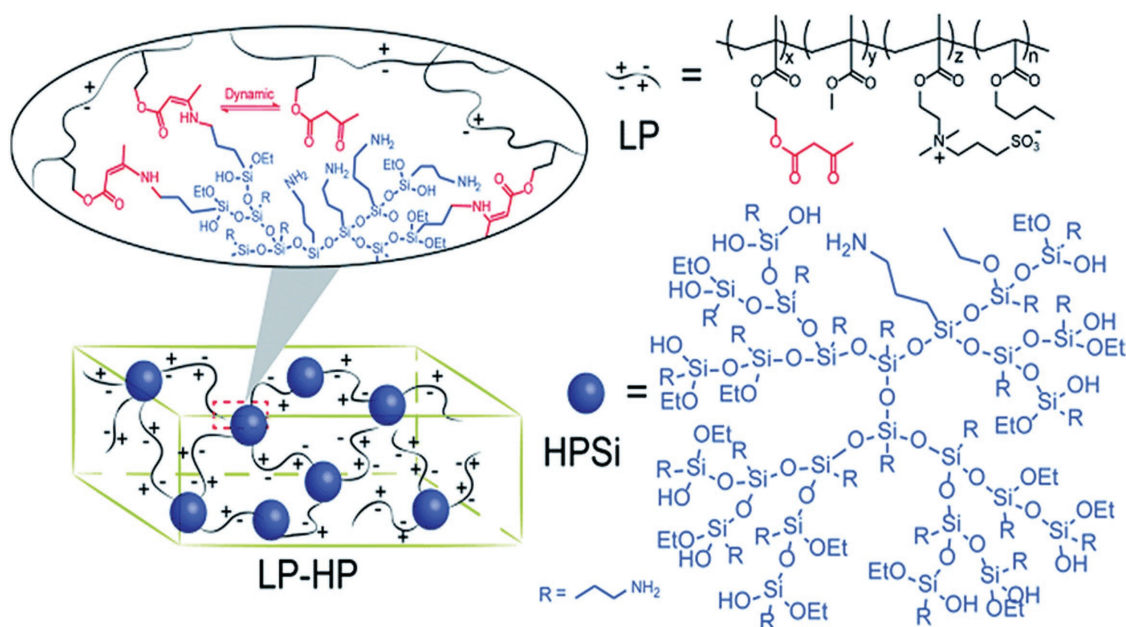


Figure 21. Structures of linear polyacrylate resin (LP), hyperbranched polysiloxane (HPSi), and LP-HP coatings. Reproduced with permission.^[66] Copyright 2017, Royal Society of Chemistry.

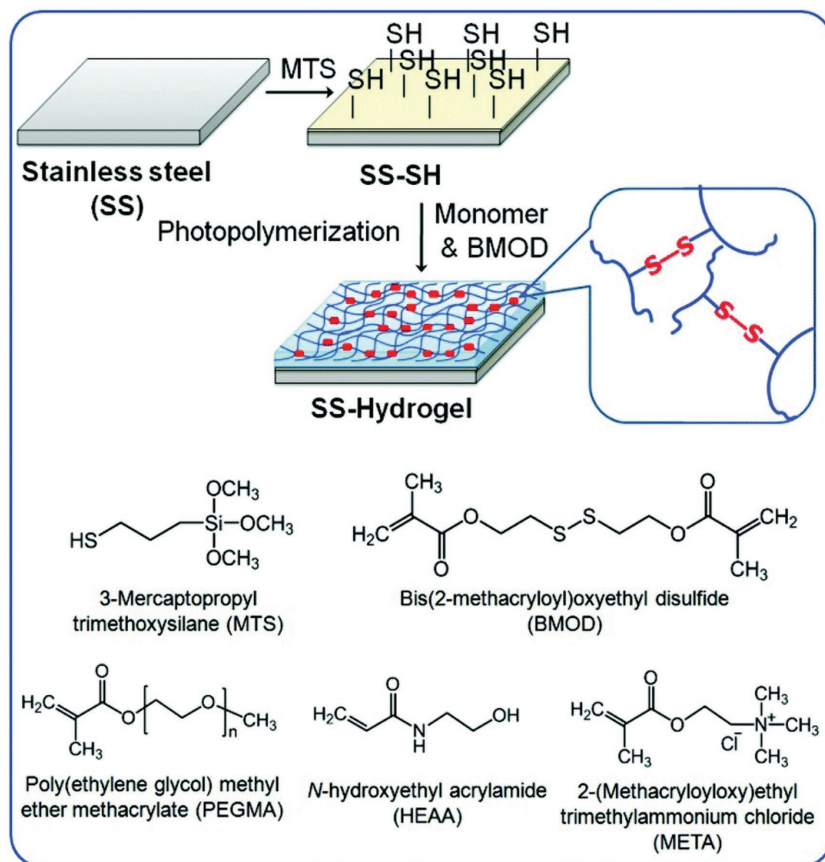


Figure 22. Schematic preparation of hydrogel functionalized stainless steel and the structure of functional monomers. Reproduced with permission.^[67] Copyright 2015, Royal Society of Chemistry.

multilayer coating with self-polishing ability.^[79] As shown in **Figure 24**, a mussel-inspired polydopamine adhesive layer was first deposited on the stainless steel surface to immobilize dextran aldehyde. Then, the antimicrobial carboxymethyl chitosan and hydrophilic dextran aldehyde were alternatively deposited via aldehyde-amine reaction. The as-prepared coatings exhibited antiadsorption of BSA and resistance of *S. aureus* and *E. coli*, and the anti-biofouling ability could be improved with increasing number of dextran aldehyde/carboxymethyl chitosan layers. Notably, because of the pH sensitivity of the imine linkage, the coating possessed self-polishing effect as imine linkage could cleavage under acidic environment. This coating showed enhanced anti-biofouling efficiencies especially under high concentrations of bacteria compared to traditional coatings.

3. Potential Use of Anti-Biofouling and Healable Materials in Biomedical and Biosensing Applications

As previously mentioned, anti-biofouling and healable materials have attracted great attention for biomedical and biosensing applications due to their structure and antifouling property integrity. Meanwhile, to extend their

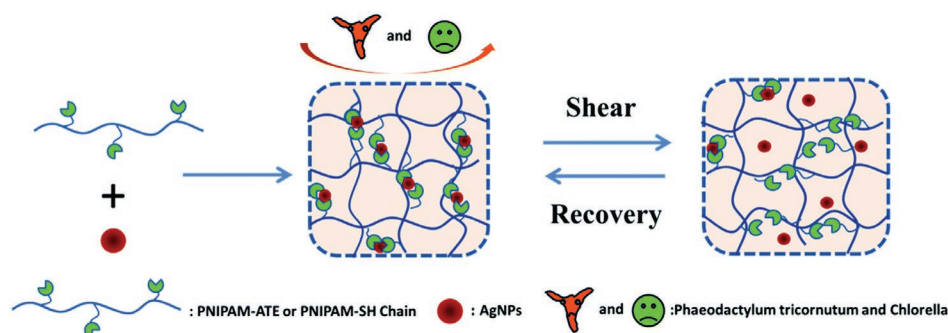


Figure 23. Schematic illustration of the preparation and self-healing process of nanocomposite hydrogels. Reproduced with permission.^[69] Copyright 2017, Royal Society of Chemistry.

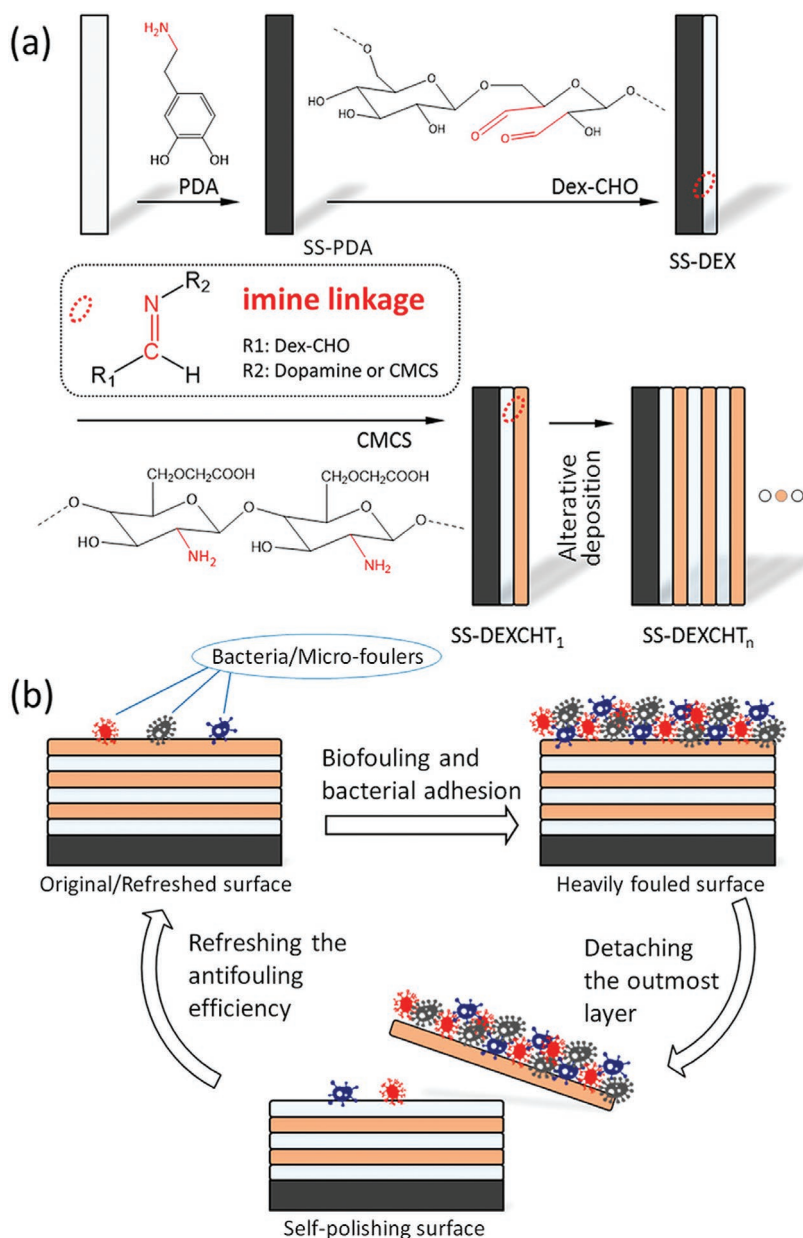


Figure 24. a) Preparation of multilayer coatings on the stainless steel surface; b) the self-polishing mechanism. All panels reproduced with permission.^[79] Copyright 2018, American Chemical Society.

potential biosensing and biomedical applications, typical factors such as distinct structure, biocompatibility, biodegradability, stimuli responsiveness, and biostability should be considered in the design and functionalization of such materials. Several representative designs of anti-biofouling and healable materials that have been used or can be potentially used in biomedical fields are listed below.

3.1. Implanted Materials

As plasma proteins and blood cells can undesirably adsorb onto the surface of implanted materials and medical devices; blood coagulation, inflammation, or bacterial infection would occur, which are the major reasons of implant failure. The development of anti-biofouling and healable surfaces can solve this problem to some extent, as summarized in the previous sections. Among them, Jiang and co-workers^[36] reported a strategy and further applied it to a catheter model. Briefly, they fabricated a catheter prototype with protonated tertiary carboxybetaine isobornyl elastomer covering the outer and inner surface via mold casting and bulk polymerization process. After hydrolysis, both the outer and inner surfaces became superhydrophilic and antifouling, while the tube remained highly elastic. This strategy avoided sophisticated surface coating process and was promising to facilitate the fabrication of any device with complicated structures.

Alginate films have been widely used in biomedical fields due to their biocompatibility, low toxicity, and more importantly, the resistance of proteins and cells.^[80] In 2016, a mussel-inspired multilayered alginate film on solid substrates was reported through spin coating of alginate catechol on a polydopamine coated surface and subsequent crosslinking via catechol- Fe^{3+} -catechol

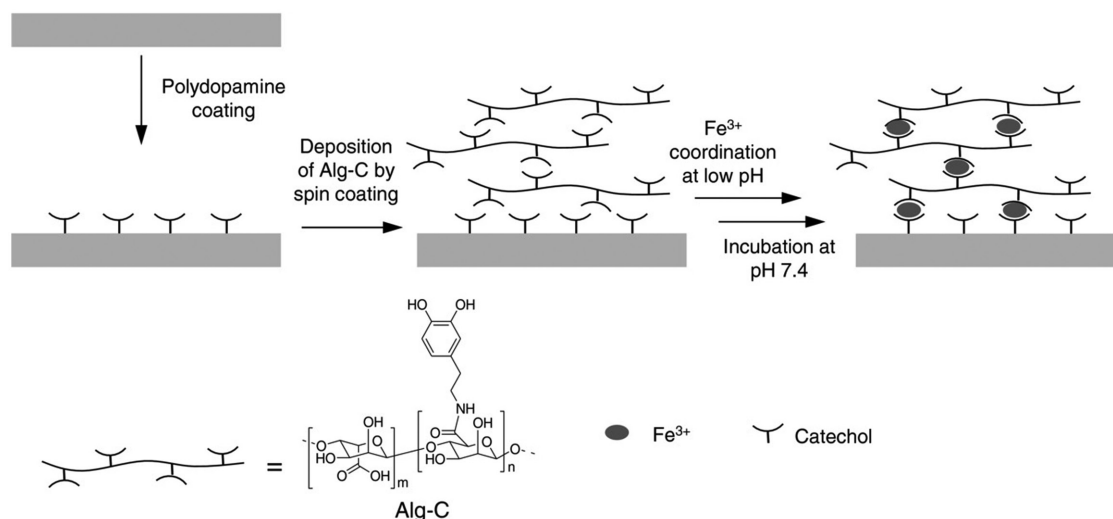


Figure 25. Schematic representation of the mussel-inspired multilayered alginate film. Reproduced with permission.^[81] Copyright 2016, Wiley-VCH.

dynamic bonds (Figure 25).^[81] This film showed excellent anti-adhesion to bacteria and antiadsorption of proteins. As the film is also biocompatible, it has the potential to be applied in the construction of functional coatings on implanted medical devices. Nevertheless, more efforts should be made on their practical applications.

Hydrogels with both antimicrobial and antifouling properties as coating on implant devices can prolong their lifetime by killing and reducing attached microorganisms. Efforts have been made in the developing of antimicrobial and antifouling hydrogels.^[82] However, the low mechanical property of the hydrogels severely hindered their applications in implanted materials. To solve this problem, Cheng and co-workers^[82b] designed zwitterionic polymer-based hydrogels that showed switchable antimicrobial ability under acidic conditions and antifouling property under neutral or basic condition. More importantly, their work successfully improved the hydrogels' mechanical property by incorporating hydrogen bond-forming hydroxyl groups, which broadened their applications as implanted materials.

3.2. Tissue Engineering

Tissue engineering involves the utilization of growth factors, cells, and scaffolds to repair a portion of tissues. The scaffolds should have good cell adhesive properties, biocompatibility, and biodegradability, as well as similar structural and mechanical properties of specific tissue. Anti-biofouling and healable materials, especially injectable self-healing hydrogels have been demonstrated to have great potential in tissue engineering. For example, the hydrogel (Figure 19a) designed by Guo and co-workers^[63] exhibited potential cardiac repair applications. First, the hydrogel could support the proliferation of the C2C12 cells and prevent them from injury when injected into the body. Second, C2C12 cells and H9c2 cells could be encapsulated in the hydrogel, and showed a controlled linear-like release profile. Furthermore, subcutaneous injection confirmed that the hydrogel has injectability and in vivo degradability. When

injury occurred, wound dressing could act as a barrier that made wound closure, promoting wound healing and reducing scar formation. Self-healing hydrogel dressings with multiple functionalities including anti-biofouling and antioxidative are highly desired in fabricating wound dressings. The hydrogels reported by Chen and co-workers^[65] (Figure 20) were able to accelerate wound healing, which was demonstrated by in vivo wound healing tests of rat skin. The wounds treated with this material showed faster contraction rates than others, and the wound closure rate of 97% was achieved 14 days after post-transplantation. Kim and co-workers reported on the synthesis of antifouling zwitterionic hydrogels impregnated with Ag NPs to produce antimicrobial activity.^[83] Both in vitro and in vivo experiments suggested that these antibacterial and antifouling zwitterionic hydrogels significantly improved the quality of the granulation tissue, vascularization, big rete pegs, and keratinocyte restoration in wounds without inflammation.^[83]

3.3. Drug Delivery

Some abovementioned anti-biofouling and healable materials are also applied to develop drug delivery systems based on their structures. For example, Kundu and co-workers^[46] prepared a self-healable polyelectrolyte film and used it to deliver curcumin, a hydrophobic drug that has antimicrobial activity and wound healing capability (Figure 12). To verify its potential application, wound healing assay was performed to observe the effect of curcumin on cell proliferation and migration, and the cell transfected with the curcumin loaded film repaired the wound predominantly compared to the pure film. In 2016, Zhou and co-workers successfully functionalized mesoporous silica drug nanocarriers with a zwitterionic anti-biofouling surface to reduce nonspecific proteins absorption and facilitate the highly selective drug delivery to tumor cells.^[84]

Injectable hydrogels are also promising materials for biomedical applications like drug delivery due to their biocompatibility, ease of operation, minimal invasion, and most importantly, similarity with natural extracellular matrices.^[85] For

instance, the injectable hydrogels reported by Zeng and co-workers^[55,56] were crosslinked by hydrogen bondings and aromatic interactions from catechol functional groups, which belonged to physical hydrogels and did not need any triggers for in situ gelation. The drug can be loaded in preformed hydrogel before injection. The catechol-mediated self-healing hydrogels greatly reduced cytotoxicity due to their metal-free nature, and attracted interests in the applications in drug delivery fields. However, more efforts still need to be made to realize their applications in in vitro and in vivo drug deliveries.

3.4. Biosensing

Label-free affinity biosensors have become important tools in the research of molecular biology and hold great potential for bioanalytical applications, such as medical diagnostics, environmental monitoring, and food safety due to their ability of direct real-time observation of interactions among biomolecules and rapid detection of chemical and biological species. Generally, these sensing systems utilize antibody or nucleic acid-derived materials (DNA/aptamer) as biorecognition elements, and the detection of analytes are based on the change of optical (reflection/absorption/fluorescence spectroscopy) or electrical (impedance/electric current) properties. However, the nonspecific fouling (primarily adsorption of proteins) of surfaces from biological media hampers the use of affinity biosensor technologies to a great degree. The fouling effects may be particularly severe during the analysis of blood plasma or serum.^[2]

To achieve low fouling for biosensing, several approaches to the fabrication of antifouling coatings have been developed, such as oligo(ethylene glycol)-terminated or zwitterionic-based self-based monolayers, PEG-based coatings, polymer brushes, which have been reviewed previously.^[2] Justin Gooding and co-workers reported for the first time that zwitterionic coatings on electrodes efficiently limited biofouling by proteins (bovine serum albumin and cytochrome c) but still maintained sufficiently low impedance to allow the proceeding of Faradaic electrochemistry. The modified electrodes were expected to be applied in electrochemical biosensors and implantable electrodes.^[86] Luo and co-workers have reported a series of antifouling electrochemical immunosensors by elaborately designing the structure of antifouling polymer-functionalized conducting materials or self-assembled antifouling peptides on the surface of electrode. These biosensors have been proved to be able to prevent the nonspecific protein adsorption onto the sensing interface even in 10% v/v human serum.^[87]

With the durability or long-term use of antifouling materials in mind, it is better to introduce the self-healing or healable abilities in biosensors. Generally, the antifouling coatings in biosensor are several tens of nanometers thick, and the foulant is mainly protein. The structural rearrangement-based antifouling mechanisms and ionic-based LbL assembly may

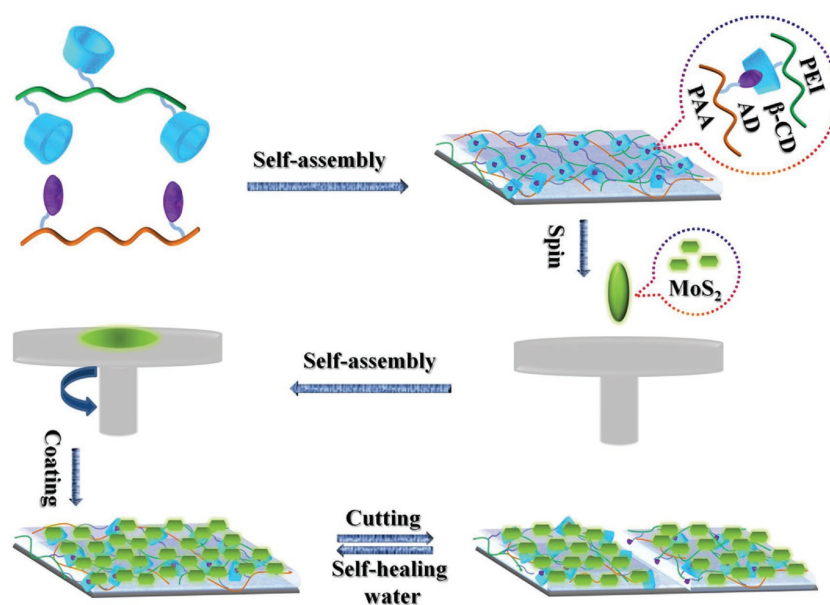


Figure 26. Schematic of the preparation of the self-healing $\text{MoS}_2/\beta\text{-CD-PEI/AD-PAA}$ antibacterial coatings via LbL assembly. Reproduced with permission.^[88] Copyright 2017, Elsevier.

have potential to prepare such kinds of coatings in the near future. For example, Ge and co-workers reported self-healing, antibacterial, and metal-ion sensing coatings via LbL assembly technique.^[88] As shown in **Figure 26**, the aqueous solutions of adamantamine (AD)-PAA and β -cyclodextrin (β -CD)-PEI were alternately deposited on the substrate to obtain the (PAA-AD/PEI- β -CD)₁₅ coatings based on host-guest interaction, and MoS_2 solution was spin-coated subsequently. The antibacterial property was endowed by the photocatalytic disinfection activity of MoS_2 nanosheets, and it could be significantly improved by UV irradiation. The SEM images showed that after scratching several tens of micrometers width, the interface of the scratch was able to swell when exposed to moisture, and the coatings self-healed through the host-guest interactions. Besides, these self-healable anti-biofouling coatings were used for detecting heavy metal ion, Co^{2+} , based on fluorescence quenching mechanism, and the detection limit was 0.018 mg mL^{-1} with a dynamic range over $0\text{--}0.1 \text{ } \mu\text{g mL}^{-1}$. This simple and easy approach to prepare the self-healing coatings with antibacterial and sensing properties offers promising opportunities in the field of biomedical and food analysis.

Electrically conductive fabrics have great potential applications in wearable biomedical sensors. However, the undesired wetting by water or organic liquids may hinder the long-term application, and even endanger the wearers if short-circuit occurs. To solve this problem, Sun and co-workers reported the preparation of electrically conductive cotton fabrics with healable superamphiphobicity recently.^[89] As shown in **Figure 27**, PdCl_4^{2-} was uniformly deposited on fabrics through LbL assembly technique and acted as catalyst for electroless deposition of Cu. The fabrics were then deposited with F-POSS and 1H,1H,2H,2H-perfluorooctyltriethoxysilane (POTS) by means of solution dipping. The obtained fabrics exhibited the abilities of self-cleaning and antiacidic/-basic solutions due to the superamphiphobicity of F-POSS^[31] and POTS. They also

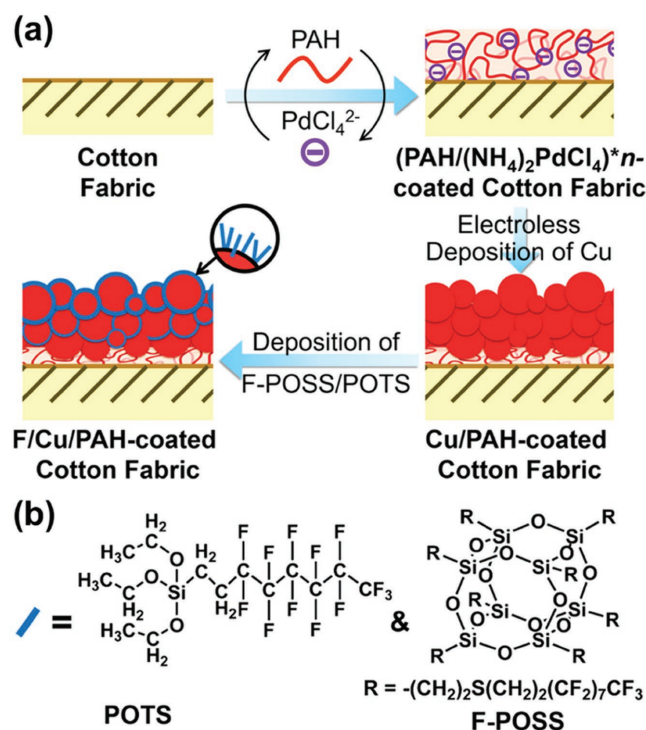


Figure 27. a) Schematic illustration of the fabrication process of electrically conductive cotton fabrics with healable superamphiphobicity; b) chemical structures of POTS and F-POSS. All panels reproduced with permission.^[89] Copyright 2018, American Chemical Society.

showed a sheet resistance of about $0.33 \, \Omega \, \text{sq}^{-1}$ and excellent electromagnetic interference shielding ability. Moreover, after being damaged, the fabrics could restore their superamphiphobicity by applying a voltage of 1.0 V or heating at $135 \, ^\circ\text{C}$ to facilitate the migration of the preserved F-POSS and POTS to the surface, which is similar to their previously reported mechanism.^[31] With the advantages of superamphiphobicity, conductivity, and self-healing ability, these fabrics have great potential applications in wearable devices, especially biomedical sensors.

4. Conclusions

In this review, we have presented some recent developments of anti-biofouling and healable materials and discussed mainly the healing mechanisms and properties of these materials. Although several excellent long-term antifouling materials have been reported, many of the works are at an early stage of application and there still is room to develop more effective materials based on innovative solutions. We believe that more attention should be paid in the following aspects when developing such materials.

First, most methods have the problem of complicated or time-consuming fabrication procedure, which limits their practical applications. Consequently, cost-effective and easy-prepared strategies should be developed. For example, a classical antifouling coating could become healable if dynamic interactions were introduced between the fillers and polymer

matrix, avoiding complex synthesis process of general intrinsic self-healing materials.

Second, it is of great significance to develop rapid, autonomous, and complete healing materials with appropriate mechanical, antifouling, and other multifunctional properties, such as good biocompatibility, adhesiveness, and optical, electrical, and/or magnetic properties for practical applications. However, the poor mechanical properties, slow healing process, and low healing efficiency are the main factors which greatly limit the extensive application of current antifouling materials. To overcome these challenges, efforts should be devoted to various fields, especially in chemistry, physics, and materials science, as well as the theoretical study of the systems.

Third, simultaneously possessing of healable, anti-biofouling, and propiate mechanical properties is the most important goal in this research field. The introduction of antifoulants and healants should not weaken the virgin mechanical properties significantly, and both mechanical properties and antifouling ability should recover as much as possible within a short healing time. Furthermore, it is better to minimize or avoid of the use of external stimuli.

Last but the most important, to design the healable anti-biofouling materials for further practical application, more specific properties should be considered in practical applications.

In summary, the fabrication of durable, healable, and anti-biofouling materials has an enough room to be explored. We hope that our personal opinions can help readers quickly catch hold of the recent progress in antifouling healable materials and develop more important fabrication methods for fundamental investigations and practical applications in the near future.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

antifouling, biosensing, implants, self-healing, tissue engineering

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